My House, My Rules: Chinese Aid, Leader Birth Regions, and the Violation of Protected Areas *

Stefano Jud† Anthony Luongo‡

Abstract

Existing research highlights that Chinese foreign aid is often vulnerable to elite capture in recipient countries. We examine whether such capture undermines environmental protection, proxied by whether aid projects are sited within protected areas (PAs). We hypothesize that leaders are more likely to authorize incursions into protected areas when projects are located in their home region, as local beneficiaries can translate the resulting rents into political support for the incumbent, while the risk of substantial public backlash over environmental violations is comparatively muted. Using geocoded data on 3,675 Chinese infrastructure projects, we find no overall increase in PA encroachment associated with leaders' birth regions. However, the aggregate null masks substantial geographic heterogeneity: in Africa, projects placed in a leader's birth region are roughly 10 percentage points more likely to encroach on PAs than projects outside that region, whereas outside Africa the relationship reverses, with home-region projects less likely to violate PAs. Additional analyses suggest that domestic public opinion toward China and the political salience of ethnicity may help account for this divergence. We find no analogous effect for World Bank projects.

*We thank Austin Beacham, Julia Gubler, Niklas Hänze, Tao Li, Quynh Nguyen, Miguel Rueda, Matt Winters, and Longjiao Yu for helpful feedback on this project. We are also grateful to audiences at the Environmental Politics and Governance Conference 2025, Midwest Political Science Association 2025, Political Economy of Aid and Development Society 2025, and the World Trade Institute (University of Bern) for their helpful suggestions.

†Email: stefano.jud@gmail.com; Postdoctoral Fellow, Wyss Academy for Nature and Institute of Political Science, University of Bern, Bern, Switzerland

‡Email: anthony.c.luongo@gmail.com; Postdoctoral Fellow, Niehaus Center for Globalization and Governance, Princeton University, Princeton, NJ, USA

1 Introduction

A major challenge with foreign aid is the disproportionate influence of elites (e.g., Andersen, Johannesen, and Rijkers 2022; Svensson 2000; Alesina and Weder 2002). This concern of elite capture appears to be particularly salient in the context of Chinese foreign aid. Empirical evidence suggests that Chinese aid is more frequently allocated to the birth regions of African leaders (Dreher et al. 2019), increases local corruption (Isaksson and Kotsadam 2018a; Brazys, Elkink, and Kelly 2017), and is more likely to fund large-scale prestige projects, such as stadiums, which fail to address the broader population's needs (Strange 2024; Wang 2022).

The susceptibility of Chinese aid to elite capture poses significant challenges for aid effectiveness. Evidence indicates that foreign aid influenced by elite interests may suffer from corruption (Burnside and Dollar 2000), poor project selection (Dreher, Eichenauer, and Gehring 2018), and low implementation rates (Williams 2017). Furthermore, it can exacerbate within-country inequalities through the misallocation of resources to regions favored by elites (Briggs 2021; Dreher, Fuchs, Parks, et al. 2021).

Elite capture in foreign aid may also have a potential impact on the environment. Conservation and environmental laws rely on effective government enforcement. However, evidence suggests that politicians often interfere with and weaken law enforcement when it serves their personal interests (Holland 2015; Sun 2021). Consequently, the prospect of rents from Chinese aid projects may diminish governments' incentives to enforce environmental laws adequately.

The aim of this paper is to investigate the extent to which elite influence contributes to Chinese aid projects posing potential harm to the environment. We specifically focus on whether Chinese infrastructure projects violate protected areas (PAs). PAs are usually located ecologically valuable areas in which economic activity is either prohibited or strongly limited (Naughton-Treves, Holland, and Brandon 2005). Anecdotal evidence suggests that Chinese aid projects are often located within or near environmentally sensitive areas, such as national parks and other protected regions (e.g., Baehr, BenYishay, and Parks 2023). For example, in Kenya, China financed the 470 km-long Standard Gauge Railway (SGR), connecting Mombasa to Nairobi. The SGR route traverses several national parks, which are designated as PAs.

We propose that when a potential Chinese project encroaches on a PA, the incumbent leader faces a political trade-off. On the one hand, infrastructure projects inside PAs can generate substantial rents for firms and politically connected elites, who benefit from construction contracts, regulatory flexibility, and expanded economic activity. On the other hand, PAs often enjoy

broad public support (Michaelsen, Sundström, and Jagers 2025), and their violation can provoke backlash, including public protest and reputational costs for the government (Bontempi et al. 2023). Leaders therefore face a balancing act. Allowing projects that harm PAs may consolidate elite support but risks alienating environmentally concerned citizens. The decision to weaken or override conservation protections is thus not purely economic, but deeply political.

We argue that leaders manage this trade-off strategically by exploiting political geography. Specifically, Chinese aid projects are more likely to violate protected areas when they are located in the incumbent's home region. In such regions, the political returns to rewarding local firms are higher, as economic gains are more likely to reinforce the incumbent's support base. At the same time, the political costs of environmental harm are lower: voters and firms in the leader's home region face a credible commitment problem in shifting their support to a challenger who may favor a different regional base. As a result, PAs are particularly vulnerable when it involves a project that benefits the incumbent's birthplace. This logic implies that elite capture of Chinese aid does not operate uniformly across space but is conditioned by the leader's regional political incentives.

Using a sample of 3,675 Chinese infrastructure projects from 2000 to 2020, we estimate the effect of a leader's birth region on the probability that a project is located in a PA. While the aggregate estimate is indistinguishable from zero, we discover significant geographic heterogeneity in our results. In the African context, Chinese aid projects placed in birth regions are significantly more likely to violate PAs. In contrast, Chinese aid projects placed outside of Africa are less likely to encroach on PAs when they benefit a leader's birth region. We find no such effect for World Bank projects in infrastructure-intensive sectors.

These results align with the scope conditions of our theoretical framework. We argue that the costs of violating a PA are lower when ethnicity is politically salient and when citizens view China relatively positively. Consistent with these expectations, Chinese aid projects in the leader's birth region are most likely to violate PAs in country-years where ethnicity is politically salient and public opinion towards China is favorable. Africa is distinct from the rest of our sample in that, on average, ethnicity is more salient and China is more popular.

This paper makes important contributions to several fields of studies. First, it further investigates the effects of Chinese aid on governance outcomes. Much has been written on how Chinese aid affects economic growth and governance outcomes (Dreher, Fuchs, Hodler, et al. 2021; Dreher, Fuchs, Parks, et al. 2021; Brazys, Elkink, and Kelly 2017; Brazys and Vadlamannati 2021; Isaksson and Kotsadam 2018a, 2018b), yet relatively few studies have touched upon the

effects of Chinese aid on environmental outcomes. Studies by Marty et al. (2019) and BenYishay et al. (2016) find that Chinese aid had mixed effects on deforestation. Dreher et al. (2022) record that Chinese aid had a mixed effect on national environmental outcomes. We expand this work on the interaction of Chinese aid with environmental politics by focusing on how Chinese aid conflicts with *de jure* efforts by governments to protect environmentally sensitive areas.

Second, it contributes to research on the efficacy of PAs for environmental conservation. Previous studies on the political economy of PAs suggest that the design and locations of PAs is strongly influenced by political considerations of the government (Mangonnet, Kopas, and Urpelainen 2022; Dawson et al. 2025; Alger 2023; Beacham 2023). This work focuses on how politics at the adoption stage affects PA effectiveness. This study looks at the enforcement stage and how politics potentially undermines PAs. Our findings indicate that, under specific conditions, PAs' ability to deter public infrastructure construction is significantly weakened when the incumbent leader hails from a region benefiting from an infrastructure project. This speaks to importance of government officials in assuring the effectiveness of PAs (Sundström 2016).

Third, we advance the study of birth region effects. Previous work theorizes that birth regions are attractive targets for aid in order to curry political favors from leaders (Dreher et al. 2019; Hodler and Raschky 2014; Briggs 2021; Bomprezzi et al. 2024b; Berlin, Desai, and Olofsgård 2023; André, Maarek, and Tapo 2025). This literature is primarily built on studies of African politics. As our results show, this focus on Africa has hidden important geographic variation. In this study, we aim to more clearly delineate the substantive conditions under which researchers should expect birth regions to shape aid allocation, which is key to understanding the extent to which birth region effects may (or may not) appear in other contexts.

2 Foreign Aid, Elite Capture, and Environmental Governance

Foreign aid is often vulnerable to elite influence in recipient countries. Donor agencies typically collaborate with local governments to implement aid projects, creating opportunities for elites to exploit incoming funds for personal gain. Research has linked foreign aid to increased corruption and rent-seeking behavior (e.g., Svensson 2000; Alesina and Weder 2002). For example, Andersen, Johannesen, and Rijkers (2022) find that countries receiving higher inflows of World Bank aid experience increased money transfers to tax havens, suggesting that elites siphon off a portion of the funds. Beyond direct financial capture, elites also shape the sub-national allocation of aid,

directing resources toward politically significant districts (Briggs 2021; Jablonski 2014; Masaki 2018).

Elite capture is also a significant concern in the context of Chinese aid. Studies have shown that corruption tends to increase in areas near Chinese-funded projects (Isaksson and Kotsadam 2018a; Brazys, Elkink, and Kelly 2017). Moreover, Chinese aid is strongly influenced by the political interests of recipient government leaders. Unlike World Bank aid, Chinese assistance is more likely to be directed toward leaders' birth regions (Dreher et al. 2019), targeted at politically strategic sectors such as infrastructure (Zeitz 2021), and provided without governance-related conditions (Li 2017). Additionally, Chinese aid can contribute to longer leadership tenures (Shea, Reinsberg, and Kern 2025).

What are the consequences of elite capture? From an economic perspective, elite capture of foreign aid can undermine aid effectiveness (Hodler 2007). For example, it has been linked to higher rates of project non-completion (Williams 2017). Additionally, as Labonne (2016) demonstrates, local political business cycles in the Philippines can hinder the long-term development of municipalities. In the context of Chinese aid, however, there is no clear evidence that elite capture directly undermines effectiveness. Dreher, Fuchs, Parks, et al. (2021) find no significant difference in aid effectiveness between leader birth regions and non-birth regions. Nevertheless, their findings suggest that elite capture contributes to greater regional inequalities.

Whether elite capture undermines environmental governance remains an open question. Numerous anecdotal cases suggest Chinese projects often violate environmental standards. In East Africa, a Chinese state-owned firm partnered with Tanzania and Uganda to build an oil pipeline through ecologically sensitive areas, including a Ramsar-protected wetland, despite international opposition (EJ Atlas 2023). In Sri Lanka, a Chinese-funded airport near wildlife sanctuaries disregarded environmental guidelines, harming local fauna (EJ Atlas 2022a). Similarly, in Ecuador, a Chinese-owned gold mine polluted waterways serving 1,500 families (EJ Atlas 2022b).

Recent studies offer mixed evidence on whether Chinese aid undermines environmental governance. In Cambodia, Chinese-funded highways have been linked to deforestation by connecting plantations and concessions (Baehr, BenYishay, and Parks 2023), while no such impact was found in Burundi and Rwanda (Marty et al. 2019). Chinese aid increased deforestation in Tanzania but reduced it in Cambodia (BenYishay et al. 2016). The environmental effects of infrastructure aid depend on project type; for example, donor-funded irrigation projects in

Cambodia have improved forest cover by easing pressure on nearby forests (Baehr, BenYishay, and Parks 2021).

An explanation for these varying results is that existing research does not fully account for the political economy dynamics driving the allocation of aid. There is strong evidence suggesting that leaders strategically implement environmental protection measures, such as the designation of protected areas (PAs), to enhance their political standing (Beacham 2023). While PAs are intended to promote conservation, they have distributional consequences, creating winners and losers, which makes them attractive targets for political manipulation. The establishment of PAs restricts or prohibits natural resource extraction, disrupting local economies that rely on these activities. Leaders can exploit these economic consequences to generate backlash against political opponents or, conversely, avoid designating PAs in areas where industrial interests are crucial to maintaining their political coalition (Mangonnet, Kopas, and Urpelainen 2022; Alger 2023; Beacham 2023; Dawson et al. 2025).

If leaders strategically place PAs, it is also plausible that they strategically enforce PA regulations, particularly when a Chinese aid project is located within a PA. There are numerous examples from other policy areas demonstrating selective law enforcement in ways that serve political interests. For instance, Holland (2015) documents how South American governments deliberately fail to enforce street vending regulations because poor voters represent key constituencies. Similarly, Mattingly (2016) shows that property expropriations in Chinese villages become more frequent when influential leaders from local social institutions are integrated into the local party cadre.

Previous research on PAs has not explored whether PA enforcement—as indicated by the placement of Chinese aid projects within PAs—is linked to elite capture. There is strong evidence that PAs can mitigate deforestation (e.g., Andam et al. 2008) and protect biodiversity (e.g., Brodie et al. 2023). However, these findings may mask heterogeneous effects, suggesting that the effectiveness of PAs depends on the absence of elite capture dynamics that allow infrastructure projects to be built within them.[1] The following section develops a theoretical framework to explain why and under what conditions elite capture can lead to PA violations, i.e., the placement of Chinese aid projects in PAs.[2]

1. There is evidence that infrastructure within PAs is harmful to the environment. While it may not directly cause deforestation, it can contribute to biodiversity loss by disrupting wildlife habitats (Xiong et al. 2025).

2. The contention, of course, is not that China is the only provider of foreign aid who funds projects that violate PAs; indeed, anecdotes abound of World Bank projects causing negative environmental outcomes. The contention is simply that China's foreign policy orientation and practices make it a most-likely case for our theory.

3 Why Violate Protected Areas?

We consider a decision-theoretic problem for a political leader who is interested in using foreign aid and investment projects to retain political power. In order to achieve this goal, the leader must appeal to both the masses and to other elites.[3] In particular, the leader must demonstrate that a sufficiently large number of each group is better off with the leader in power than they would be under the rule of some challenger. Masses can vote against the leader or take to the streets if unhappy, while elites can withhold campaign contributions and endorse challengers. A leader's main goal is minimize these threats.

One tool leaders use to secure political support is foreign aid. Foreign aid allows politicians to claim credit which allows them to improve their political standing among the public (e.g., Cruz and Schneider 2017; Baldwin and Winters 2023; Dietrich, Mahmud, and Winters 2018; Springman 2022). They can also influence where aid projects are allocated (e.g., strongholds or swing districts) and reap political gains from the distributive effects of aid (Anaxagorou, Efthyvoulou, and Sarantides 2020; Jablonski 2014; Masaki 2018; Hodler and Raschky 2014). Foreign aid is also an opportunity for elites to extract rents from these projects, for instance, by receiving a contract from an aid project (e.g., Andersen, Johannesen, and Rijkers 2022; Bjørnskov 2010). In addition, foreign aid can improve the local business environment and improve local firm profitability (Marchesi, Masi, and Paul 2025). Given these political benefits, there is evidence that politicians pursue aid projects strategically prior to elections (Chen 2025; Kersting and Kilby 2016) and that foreign aid can extend a leader's tenure (e.g., Shea, Reinsberg, and Kern 2025; Licht 2010; Yuichi Kono and Montinola 2009).

Chinese aid is particularly vulnerable to be captured by leaders in recipient countries for their political motives. In general, recipients are better able to shape Chinese aid in their favor as Chinese aid projects have fewer strings attached (Dreher et al. 2022). This gives recipients the chance to design domestic policies in line with their preferences (Kaplan 2021) and allocate funding projects in their preferred areas (e.g., the leader's home region) (Dreher et al. 2019; Dreher, Fuchs, Parks, et al. 2021; Bomprezzi et al. 2024b). Furthermore, Chinese aid projects tend to be less transparent, which means it is harder to observe for outside actors and to hold the government accountable (Brown 2025; Cormier 2023). All these features make Chinese aid attractive for a leaders to extend their tenure (Shea, Reinsberg, and Kern 2025).

3. Elites encompasses political, social, military, and economic leaders. We focus specifically on the role of firms, who are important actors in both democracies and autocracies.

A key challenge for political leaders is that some Chinese aid and infrastructure projects, if implemented, would violate the boundaries of PAs. Numerous cases illustrate how Chinese-financed infrastructure has generated controversy due to its location within environmentally sensitive areas. For example, in the Democratic Republic of the Congo (DRC), the government proposed the construction of the USD 500 million Sombwe hydropower dam in collaboration with PowerChina. The project is planned inside Upemba National Park, one of the country's most important protected areas, and has drawn criticism from environmental NGOs as well as international actors. The European Union's ambassador to the DRC publicly questioned the legality of constructing the dam within a national park (Global Witness 2021). A similar case is the Olkaria geothermal power complex in Kenya, which has received Chinese financing.[4] The plants are located within Hell's Gate National Park and have raised concerns about noise pollution and gas emissions that may threaten local wildlife (Welz 2019).

These cases illustrate how economic development objectives can directly clash with conservation commitments. At the same time, important differences exist across projects. Some infrastructure projects, such as the Sombwe Dam, are not inherently location-bound and could in principle be relocated outside a protected area. In such instances, siting the project within a PA reflects a choice rather than a technical necessity. Other projects, however, are more geographically constrained. For example, geothermal power plants like the Olkaria complex must be built where underground heat resources are accessible, limiting the scope for relocation. Nevertheless, even in these cases, developers may regard locations within protected areas as economically optimal due to lower land acquisition costs, proximity to complementary infrastructure, or resource accessibility. Ultimately, it is the political leader who decides whether to approve or reject projects that encroach upon protected areas. The final determination therefore reflects not only technical feasibility but also political incentives.

In cases where an aid project intrudes environmental sensitive areas, like a PA, leaders have to deal with competing political pressures. On the one hand, economic elites are likely to favor projects that encroach upon PAs. Development within protected areas can create lucrative opportunities for firms seeking to maximize profits and may involve relaxing environmental regulations that would otherwise constrain commercial activity. This logic aligns with research on the political economy of protected area placement. Studies show that strong commercial interests in a region shape both the likelihood that a PA is designated (Beacham 2023) and the strictness

4. See https://china.aiddata.org/projects/58350/ (last accessed February 20, 2026).

of its protection (Alger 2023). Moreover, Mangonnet, Kopas, and Urpelainen (2022) demonstrate that in Brazil, protected areas have sometimes been strategically placed in opposition strongholds, potentially to undermine local economic growth.

On the other hand, citizens on average appear more supportive of conservation and may oppose violations of PAs. Public opinion research consistently finds broad backing for protected areas (e.g., Newmark et al. 1993; Michaelsen, Sundström, and Jagers 2025). A large-scale survey conducted in eight countries shows that more than 80% of respondents support the "30-by-30" target adopted in the 2022 Global Biodiversity Framework, which aims to protect 30% of land and water areas by 2030 (Michaelsen, Sundström, and Jagers 2025). Although economic investments could, in principle, offset environmental concerns, evidence suggests that citizens are unwilling to trade environmental protection for economic gains without limits. Individuals tend to reject investments perceived as environmentally irresponsible, even when they promise economic benefits (Nguyen and Malesky 2021; Moehlecke, Fasolin, and Spektor 2025; Beesley and Slaski 2024). Beyond attitudinal support for conservation, violations of protected areas have also been shown to trigger political mobilization, including public protests (Bontempi et al. 2023).

Taken together, this creates a political dilemma: satisfying economic elites by allowing development in protected areas risks alienating environmentally concerned citizens, while upholding conservation commitments may frustrate powerful commercial interests. Leaders therefore face a strategic choice in balancing these competing constituencies.

How do leaders manage this tradeoff? The political geography of the situation is key. We argue that leaders are more likely to approve projects that violate PAs when they are located in the leader's home region. Home region firms and voters are less likely to support challengers because a challenger from another region may favor their own home region. This means that the incumbent's returns to helping home region firms are higher; new profits are more likely to be contributed to the incumbent's campaign than to a challenger's. It also means that the costs of defying home region voters are lower. Home region voters face a credible commitment problem in punishing the incumbent by supporting a challenger, as they fear doing even worse under a challenger's regime with a different power base (e.g., Kasara 2007). Our argument – that the benefits of helping home region firms are high and the costs of defying birth region voters are low – generates our main hypothesis.

Hypothesis 1 (H1) *Chinese aid projects located in the incumbent's home region are more likely to be placed within the borders of a PA than aid projects in non-home regions.*

3.1 Scope Conditions

Our argument is most likely to hold under certain conditions. First, placing an aid project in a leader's home region is most likely to boost the benefits and reduce the costs of violating PAs when ethnicity is politically salient. If ethnicity is not politically salient, firms in the leader's home region are less afraid to use their new profits to back a challenger than if ethnicity were salient. Similarly, if ethnicity is not salient, voters do not face as acute a commitment problem in threatening to back a challenger, as it is less clear that they will be worse off under the challenger. The salience of ethnicity in this context is also highlighted by Dawson et al. (2025). They show that, across Africa, protected areas are less likely to be designated in regions inhabited by ethnic groups represented in the national government. Moreover, when such regions do receive a protected area, these areas are more likely to permit tourism and other economic activities, suggesting weaker or more flexible forms of protection. This discussion gives rise to a second hypothesis.

Hypothesis 2 (H2) *The marginal effect of location within the incumbent' home region on the likelihood of a project being placed within the borders of a PA increases as the political salience of ethnicity increases.*

Second, we expect that voters are more likely to punish a PA-violating leader if they hold a negative opinion of the donor (in our case, China). Recall that a politician must convince voters that they are better off than they would be under a challenger, and that this is particularly unlikely to be true in the home region. A PA violation on its own may be insufficient to cause home region voters to punish the leader, but if it comes from a particular unpopular donor, the challenger may indeed become more appealing than the incumbent. Anecdotes abound of Chinese projects engendering backlash, including in states where the government cooperates with China and when environmental degradation is a risk. In Myanmar – a longtime Chinese partner – popular protests targeted a set of investments by Chinese state-owned enterprises in dams, mines, and oil pipelines. Among the major grievances, according to Miller (2019), were environmental degradation and the exploitation of natural resources. In response to mounting public pressure, the Burmese president ultimately canceled the projects. This forms our final hypothesis.

Hypothesis 3 (H3) *The marginal effect of location within the incumbent' home region on the likelihood of a project being placed within the borders of a PA increases as the public's favorability towards China increases.*

Another common factor to plausibly explain variation in the effect of protected areas on Chinese aid placement is democracy (Beacham 2023). The importance of democracy as a moderator is unclear. On the one hand, leaders in autocracies need not be as concerned about backlash from voters; elections, if they exist, are not faithful translations of the popular will and meaningful collective action (i.e., protests) are more difficult for the masses to achieve than voting against an unpopular leader. This would imply that the marginal effect of home region decreases as a regime becomes more democratic because violating PAs is less costly in autocracies. On the other hand, leaders in autocracies may not need to use aid projects to benefit home region firms. They may use more direct methods – like bribes or preferential access to government contracts – that would be unpopular in democracies. If this effect dominates, the marginal effect of home region may not change or would increase in democracies. We remain agnostic on the importance of democracy as a moderator, but include this analysis given the importance of domestic institutions to our theory.

4 Research Design

4.1 Sample

Because our theory operates at the project level, the unit of analysis is the individual aid project rather than a time-series cross-sectional unit such as an ADM1 region-year, as used in related studies. The sample includes all Chinese infrastructure projects between 2000 and 2020 for which we have a recorded commitment year and that are implemented in a country for which we know the leader's birth region. We focus on infrastructure projects as these tend to be the most damaging to PAs due to the associated construction activities. According to the Global Chinese Development Finance (GCDF) Dataset (Goodman et al. 2024), 3,675 projects meet these criteria.[5] The sample encompasses projects valued at 615.3 billion US dollars, implemented across 118 countries and covering all types sectors ranging from eduction to mining.[6]

5. Overall, the dataset includes 9,404 projects. Of these, 5,444 were excluded because they are not classified as infrastructure projects. The remaining projects were excluded either due to missing country codes or incomplete commitment year information, which represent a minority of cases.

6. 33.4% of projects are social development projects, 20.2% transport projects, 18.9% energy projects, 11.8% education projects, and 11.7% mining projects.

4.2 Treatment and Moderators

Our main treatment variable is a binary indicator set to 1 if a project is implemented in a province (ADM1 level) that corresponds to the birth region of the current leader. The GCDF dataset allows us to precisely identify all ADM1 provinces where a given project is implemented. Since many projects span multiple provinces, the variable is also coded as 1 if any of these provinces include the leader's birth region. To determine whether the provinces where projects are implemented align with the leader's birth region, we utilize the newly released Political Leaders' Affiliations Dataset (PLAD) (Bomprezzi et al. 2024a).

To evaluate hypothesis 2, we interact this home region dummy with country-level variation on the level of ethnic political voting. To measure ethnic voting, we use data compiled by Juon (2024), which combines and standardizes survey questions on support for ethnically based political parties among 900,000 respondents in 132 countries. We aggregate this data at the country-year level, using the share of respondents in a given year who express a preference for ethnic voting as our moderating variable. For some country-years, survey data is unavailable. In these cases, we project the most recent available observation forward.[7]

For hypothesis 3, we interact a country's public opinion towards China with the home region dummy. Measuring public opinion toward China on a global and temporal scale is challenging. The most comprehensive available dataset is the one compiled by Chubb (2024), which aggregates surveys on public sentiment toward China. We focus specifically on polls that measure individuals' favorability toward China, using net favorability as our key metric. In country-years with multiple surveys, we calculate the average net favorability across polls, weighting each poll by its sample size.

However, survey data is sparse for many countries, often available only at intervals of three to four years. Given this substantial missingness, assuming that public opinion remains constant between survey years is not reasonable. To address this, we imputed the missing values using Amelia II, a software designed for multiple imputation of time-series cross-sectional data (Honaker, King, and Blackwell 2011). For the imputation, we constructed a country-year panel dataset including all countries with at least one survey observation between 2000 and 2020. We then

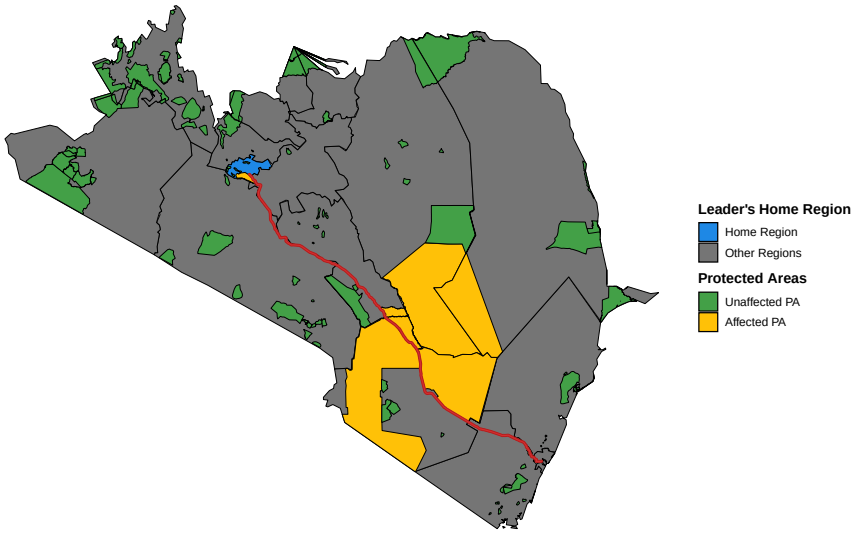
7. We acknowledge that this approach relies on the strong assumption that ethnic voting patterns remain unchanged since the last available survey. However, given that ethnic voting is often a deeply entrenched and stable feature in certain societies, we believe this assumption is reasonable.

matched this data with a set of covariates likely to influence public opinion toward China.[8] To ensure robustness, we ran ten imputation models and aggregated the results by calculating the mean imputed public opinion per country-year, aligning it with our project-level Chinese aid dataset.

4.3 Dependent Variable

To construct our dependent variable, we match geo-location data on the presence of PAs with the locations of Chinese aid projects. We retrieve the geo-locations of all PAs from the World Database on PAs (UNEP-WCMC 2019). A key advantage of the GCDF dataset is that it provides geo-coded data for the precise extent of each Chinese aid project. For example, Figure 1 illustrates the Kenya SGR project, where the GCDF dataset details the exact route of the project, depicted in red. This enables us to accurately match Chinese aid projects with PAs.

Figure 1: Kenya Standard Gauge Railway and Violations of PAs



Note: Figure shows a subset of counties in Kenya. The county colored in blue is the birth region of the incumbent leader Uhuru Kenyatta. The red line in the represents the route of the SGR. Yellow areas area PAs affected by the SGR and green areas are unaffected PAs in the map area.

There are four different ways how Chinese projects and PAs relate to each other (see Table 1). The first category comprises projects that are not located within a PA, representing the majority of cases. In fact, 78.3% of Chinese aid projects do not violate PAs. While this suggests that most Chinese projects do not directly threaten PAs, it is concerning that 21.7%—or one-fifth—of all

8. The covariates included in the imputation are Agreement with China in UN, Trade/GDP (in %), log GDP per capita, oil rents, log population, rents from natural resources, log ad from US, log aid from China, VDEM electoral democracy index, US ally, China ally, communist country, log distance to China, member of AIIB.

Chinese projects do harm PAs, endangering some of the most sensitive environments globally.[9] This issue is further exacerbated by the fact that the most destructive types of projects—transport and energy—have an even higher share within PAs, with 26.8% of transport projects and 24.9% of energy projects located inside PAs.

The second category represents projects which are located within PAs that are both inside and outside a leader's birth region. The Kenya SGR project, shown in Figure 1, serves as a prime example. At the time the SGR was signed, Kenya's leader was Uhuru Kenyatta, whose birth region is the capital, Nairobi. The SGR project encroaches on a PA within Nairobi and also passes through two PAs in counties outside of Nairobi. In our sample, 4% of all projects violate PAs both inside and outside a leader's birth region.

Table 1: Classification of PA violations

	PA Outside BR = 1	**PA Outside BR = 0**
PA Inside BR = 1	Both (N=148, 4%)	Only Inside (N=28, 0.8%)
PA Inside BR = 0	Only Outside (N=621, 16.9%)	In no PA (N=2878, 78.3%)

Note: PA stands for PA and BR is the abbreviation for birth region. Percentages in parentheses represent the percentage of cases in our data set that fall under a given category.

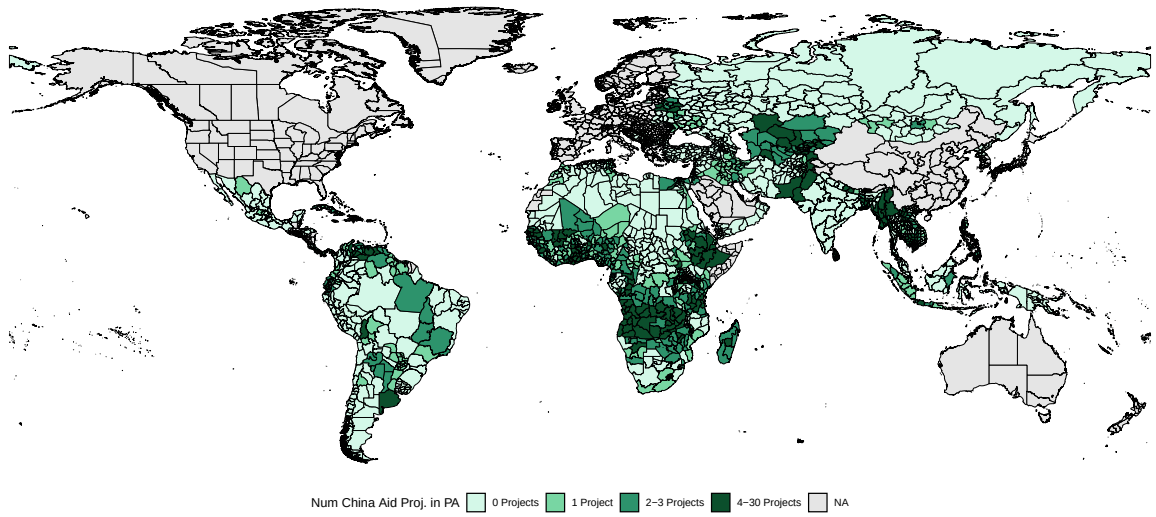
The final two categories distinguish between projects that violate PAs located exclusively outside or inside a leader's birth region. The least common scenario in our sample is when a project infringes upon a PA solely within the leader's birth region, accounting for only 0.8% of all Chinese aid projects. Conversely, the second most common scenario involves projects that violate PAs only outside the leader's birth region, representing 16.9% of the sample.

For our main analysis, we use a binary dependent variable called *In PA*. It is coded as 1 if a project violates any PA, regardless of whether the affected area is inside or outside the leader's birth region. Figure 2 illustrates the geographic distribution of Chinese aid projects that violated PAs between 2000 and 2020, dis-aggregated by ADM1 region. The figure reveals that violations of PAs are not concentrated in any specific region. Numerous Chinese projects infringe on PAs across Southeast Asia, particularly in countries like Cambodia, Myanmar, and Laos. Similarly, there are significant violations in Africa, with Angola, Ethiopia, and Ghana recording a high number of such

9. This is a conservative estimate, as one could argue that projects in close proximity to PAs also pose a threat. When we apply a 1 km and 5 km buffer around Chinese aid projects, the percentage of projects affecting PAs increases to 27.6% and 42.8%, respectively.

incidents.

Figure 2: Number of Chinese Aid Projects in a PA at ADM1 Level



Note: The figure illustrates the number of Chinese aid projects that impacted PAs within each ADM1 unit between 2000 and 2020. Grey regions indicate countries that have not received any Chinese aid project between 2000 to 2020.

Figure 3 shows the temporal development of Chinese aid projects that overlap with PAs. Overall, the share of projects placed within PAs has remained relatively stable, with a slight upward trend compared to the early years of Chinese development assistance. This pattern suggests that there has been no significant structural shift in how Chinese aid projects are sited with respect to environmental protections. Furthermore, the trend does not indicate a clear learning process—there is no evidence that earlier projects were less environmentally conscious and that Chinese actors have become more attentive to environmental concerns over time.

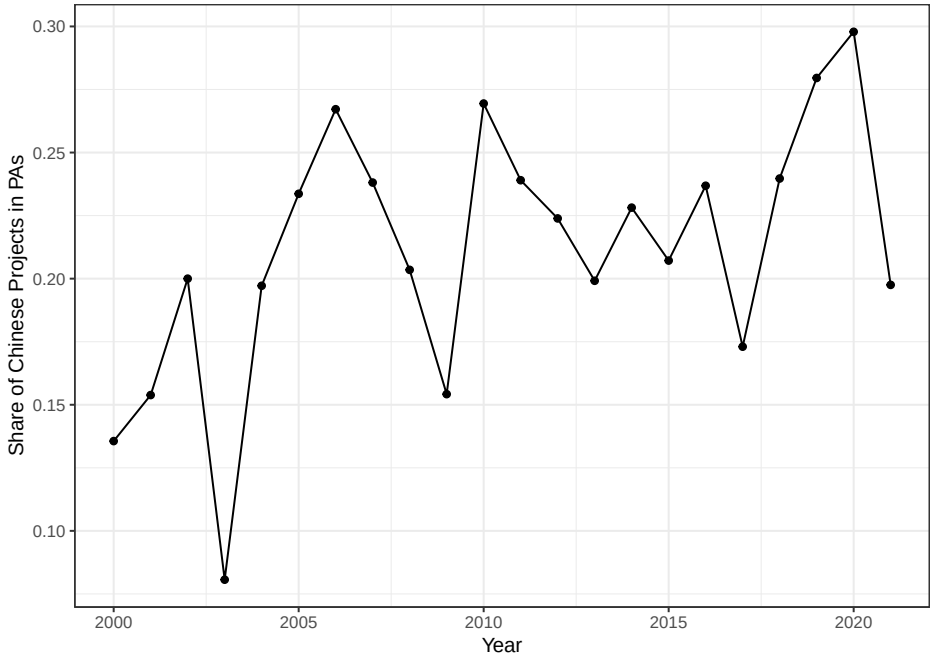
4.4 Estimation Strategy

We estimate the impact of birth region on the allocation of Chinese aid projects within PAs using the linear probability model specified in Equation 1 where i denotes an individual project, c the ADM1 regions covered by the project, t the project's start year, and s the project's country. This model specification is similar to the approach employed by Dreher et al. (2019).[10]

$$\text{In PA}_{icst} = \beta \text{Birth Region}_{ct} + \delta \mathbf{X}'_c + \lambda \text{Sector}_i + \gamma \text{Num. Regions}_i + \zeta_{st} + \epsilon_{icst} \tag{1}$$

10. The main difference is that Dreher et al. (2019) use a panel data set and not a project cross-section.

Figure 3: Share of Projects in PAs over Time



Since the allocation of projects to birth regions is not a random process, our estimation strategy employs a selection-on-observables approach. To control for country-specific confounders, we include country-year fixed effects (ζ_{st}). To address potential sub-national confounders, we incorporate a set of time-invariant control variables at the ADM1 level, denoted by $\mathbf{X}'_c$. These variables include a region's population in 2000, the total number of PAs in 2000, the total number of mines, the total number of oil and gas fields, average nighttime light in 2000, the total area in square kilometers, and whether a project is implemented in a capital region.[11] If a project spans multiple ADM1 regions, we aggregate these variables by taking the mean to create a single observation per project. Finally, we account for project-level characteristics by including sector fixed effects (λ) and controlling for the total number of ADM1 regions a project covers (γ).

Besides the non-random assignment of birth regions, a major challenge for our design is the risk of post-treatment bias. Our sample consists only of projects that have either been realized or have a firm commitment for implementation. However, this sample may be influenced by the presence of birth regions. Dreher et al. (2019) demonstrate that birth regions tend to receive more Chinese aid. Similarly, Briggs (2021) finds in a conjoint experiment with development practitioners that governments are more likely to favor projects in the home region of their leader. Therefore, it is reasonable to assume that by conditioning on project implementation, we may inadvertently introduce post-treatment bias into our estimate. To address this issue, we employ a

11. For population, nighttime light, and total area, we take the natural logarithm to normalize the distribution.

strategy proposed by Knox, Lowe, and Mummolo (2020) which allows us to assess the sensitivity of the results to post-treatment bias.

We use cluster-robust standard errors at the recipient country level. Given the likelihood of within-country spillovers in the placement of aid projects, it is reasonable to assume that observations within a country are not independent of one another. In contrast, we are less concerned about spatial spillovers across countries, as our dataset does not include cross-border projects, and project allocation decisions are primarily made by the recipient governments. For these reasons, we do not apply standard error adjustments that account for spatial proximity across countries.

5 Leader Birth Region and PA Violations

We discuss the results of our analysis in four steps. First, we present the main results. Then we discuss a placebo test using World Bank projects. Third, we explore the role of sectoral differences across projects. Lastly, we analyze which types of PAs are violated by the Chinese projects.

5.1 Main Results

Table 2 displays the effects of a leader's birth region on the placement of Chinese aid projects within PAs (see columns 1, 3, and 5). When examining the entire sample, the presence of a Chinese aid project in a leader's birth region has no significant effect on the likelihood that the project is located in a PA. In fact, the coefficient is nearly zero, suggesting that leaders' decision to respect PA boundaries is unrelated to their birth region. This outcome does not align with our theorized effect as proposed in hypothesis 1.

We then decided to differentiate between projects in African and outside of Africa. This was motivated by two reasons. First, Africa is more likely to satisfy the scope conditions we outlined in the theory section. African countries express a much higher propensity for ethnic voting than non-African countries.[12] Similarly, African countries had an average net favorability toward China of 40.6 percentage points, while outside of Africa, this net favorability was lower, at 31 percentage points. Second, previous work on leader birth regions focuses mostly on Africa (e.g., Dreher et al. 2019).

12. The average ethnic political voting score for projects in Africa is 0.6 and outside of Africa it is 0.19.The ethnic political voting score ranges between 0 and 1 (see Juon 2024).

Table 2: Effect of Birth Region on Placement in PA (Chinese and World Bank Aid)

	All		Africa		No Africa	
Model:	Chinese Aid (i)	World Bank (ii)	Chinese Aid (iii)	World Bank (iv)	Chinese Aid (v)	World Bank (vi)
Birth Region	0.006	0.015	0.099*	0.025	-0.114*	0.002
	(0.041)	(0.044)	(0.039)	(0.083)	(0.053)	(0.052)
Fixed-effects						
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes		Yes		Yes	
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
R^2	0.580	0.756	0.607	0.790	0.584	0.734
Observations	3,675	2,117	1,850	774	1,825	1,343

Note: The models include the following covariates: Number of ADM1 regions covered by project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, and number of PAs. The models also include recipient-year and project sector fixed effects. The World Bank models do not include sector fixed effects as they usually apply to more than one sector. Signif. Codes: **: 0.01, *: 0.05, +: 0.1

The results from this subset reveal a different pattern. Specifically, in African countries, Chinese projects that benefit a leader's home region are 9.9 percentage points more likely to be located in a PA compared to projects placed outside a leader's birth region. Considering that the baseline probability of a Chinese project being located in a PA is 21.7%, this effect represents a 46% increase in the likelihood that a project violates a PA.

Third, when we exclude all projects implemented in Africa from the sample, we observe an opposite pattern. In countries outside of Africa, projects targeting a leader's birth region are less likely to be placed in PAs. Specifically, the probability of violating a PA decreases by 11.4 percentage points when the project is located in a birth region. This decline is statistically significant at the 95% confidence level. These findings suggest that the aggregate null result is not genuinely indicative of a null effect but rather masks opposing dynamics across different geographic regions.

To gauge robustness of our findings, we create buffer zones around the included Chinese projects. Projects can have spillover effects that may damage PAs even when they are placed outside of them; for instance, chemicals used in industrial processes might enter waterways that flow into PAs, or trains traveling along a new railroad might release air and noise pollution that disrupt nearby ecosystems.

The results in Table B.1 indicate that the null results at the aggregate level remain consistent

even when considering projects that are just outside a PA. In the Africa sample, the effect of birth regions on the placement of projects within PAs persists under the 1 km buffer zone specification but disappears when a 5 km buffer zone is applied. While the coefficient remains positive, it is less precisely estimated, which is expected given that the 5 km buffer likely introduces more noise—i.e., projects that are not particularly contentious and do not require strong backing from the leader. For the non-Africa sample, the coefficient also remains negative but is no longer statistically significant. We attribute this change to the increase in noise as well. Overall, the results suggest that the projects most affected by the birth region effect are those that actually cross into PAs.

How robust are our estimates from a statistical perspective? We focus on two main sources of potential bias. The first concern is omitted variable bias. Descriptively, projects implemented in a leader's birth region differ systematically from those located elsewhere: they tend to be situated in economically and politically more important areas (see Tables C.1 and C.2). Although we address this concern by including a comprehensive set of pre-treatment regional covariates, the possibility of unobserved confounders cannot be fully ruled out. The direction of any resulting bias depends on whether governments are systematically more willing to violate protected areas in economically valuable regions. Theoretically, arguments can be made for both upward and downward bias. Empirically, however, Tables A.1 to A.3 suggest that our baseline estimates may be somewhat upward biased, as the inclusion of economic controls generally reduces the magnitude of the birth-region coefficient (though not uniformly across specifications).

A second concern is post-treatment bias. If some projects are strategically located in the leader's birth region precisely because of political favoritism, conditioning on certain post-treatment variables could attenuate the estimated effect. To address this issue, we implement the bounding approach proposed by Knox, Lowe, and Mummolo (2020). The results, presented in Figure D.1 in Section D of the supplementary material, indicate that our main estimates are likely conservative. Under plausible assumptions about the extent to which projects are deliberately allocated to benefit the leader's home region, the estimated effect of birth region on PA violations increases rather than decreases. Even in the African subsample—where prior research suggests post-treatment bias is most likely—the lower bound of the treatment effect remains above the estimate obtained under the assumption of no post-treatment bias.

Taken together, the overall direction of bias is theoretically ambiguous. Omitted variable bias and post-treatment bias may offset one another or reinforce each other, and either could dominate in magnitude. Importantly, our estimation strategy does not mechanically favor our preferred

interpretation.

5.2 Placebo Test: World Bank Projects

Our argument rests on the idea that Chinese aid is responsive to political leaders' birth regions due to its connection with rent-seeking behavior. To further support this claim, we conduct a placebo test by examining whether World Bank projects are also more likely to be located in birth regions and placed within PAs. Prior research shows that World Bank project allocation is not influenced by birth regions (Dreher et al. 2019) and is generally less susceptible to corruption (Brazys, Elkink, and Kelly 2017; Isaksson and Kotsadam 2018a), owing to its more stringent appraisal procedures, which include environmental safeguards (Buntaine 2015). Accordingly, we expect that birth regions should not affect the probability that a World Bank project is located within a PA.

To test this, we use the Geocoded Official Development Aid Dataset (GODAD) to identify all infrastructure-related World Bank projects between 2000 and 2021 and apply the same spatial matching strategy used for the Chinese aid analysis.[13] This results in a dataset of 2,123 World Bank projects over that period.

Surprisingly, we find that 32.2% of these projects are located within PAs—a higher share than for Chinese aid projects. However, if a project violates a PA, World Bank projects are less likely to encroach on the most strictly protected categories: only 3.3% intersect with high-stringency PAs, compared to 11% for Chinese-funded projects. This suggests that World Bank projects are more likely to be in line with the goals of a PA and advance projects that are not as harmful to the environment. Moreover, it is important to note that this figure may overestimate the problem in the World Bank case, as we cannot definitively identify which projects involve actual infrastructure components.

In line with our expectations, Table 2 shows that a leader's birth region is not significantly associated with the likelihood that a World Bank project is located within a PA (see columns 2, 4, and 6). While the coefficients across all samples are positive, none are statistically significant at the 95% confidence level. This suggests that, although a higher share of World Bank projects are placed inside PAs overall, this pattern does not appear to be driven by leaders' efforts to channel

13. GODAD does not provide a clear indicator for whether a project is infrastructure-related. We follow the approach of Zeitz (2021), who classified World Bank projects as infrastructure-related if they fell into one of the following sectors: water supply and sanitation, transport and storage, communications, energy, agriculture, forestry, fishing, industry, mining, construction, and other social infrastructure.

resources toward their birth regions for rent extraction. These findings are consistent with previous research and further strengthen our confidence in the interpretation of the China aid results.

5.3 Sectoral Differences

Another issue is that our main results plausibly depend on the sector in which projects are implemented. As Figure 1 illustrates for railways, certain transport infrastructures have virtually no routing discretion and therefore tend to cut through protected areas by construction. Moreover, transport projects generate comparatively diffuse, system-wide benefits, whereas energy and mining investments create highly localized rents. These technological and distributive features make sector-specific heterogeneity in our estimates likely.

Figure 4 shows that this expectation is borne out in the data. The birth–region effect is concentrated in projects within the energy & mining sector, while the corresponding coefficients for transport and social sectors are either negative and statistically indistinguishable from zero.[14] In other words, the aggregate pattern we document does not generalize uniformly across sectors; it is primarily driven by extractive and energy projects.[15]

This sectoral profile aligns with our broader interpretation that birth regions matter where business interests and localized economic benefits are at stake. Consistent with that narrative, we replicate the main result only for energy and mining projects, the very domains in which place-bound rents are most salient, whereas sectors with diffuse benefits or strong technological routing constraints do not exhibit the same pattern.

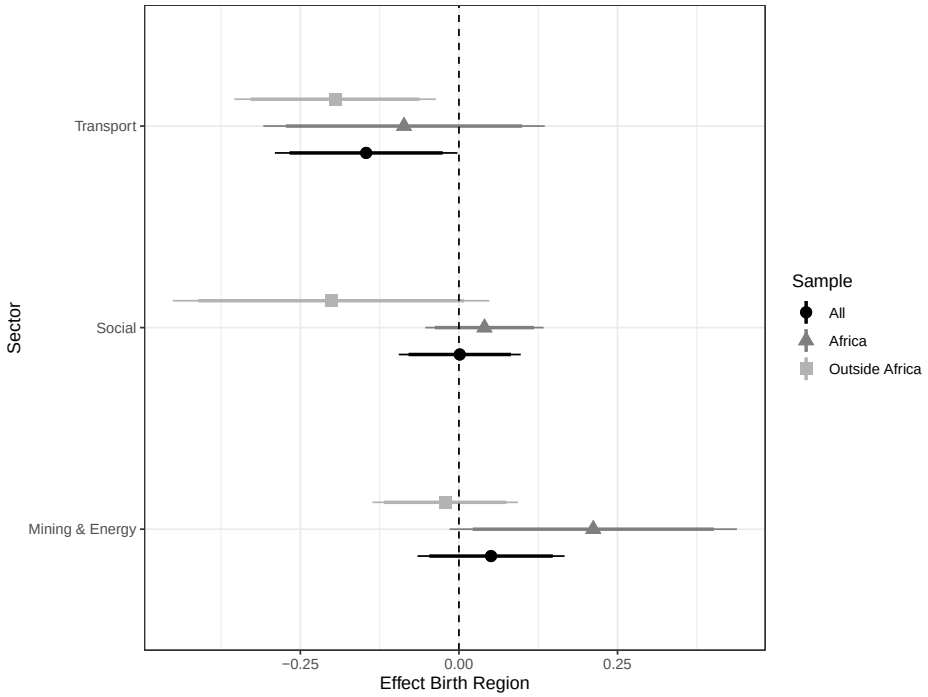
5.4 Types of Protected Areas

Another approach to better understand the results is to examine the types of PAs that Chinese aid projects encroach upon. The International Union for Conservation of Nature (IUCN) classifies PAs into seven categories, each differing in the level of permitted human activity. For example, IUCN Category Ia designates "strict nature reserves", where all human activities, except for scientific research, are prohibited. In contrast, IUCN Category VI allows for some form of sustainable

14. There are more than these sectors in the data. We chose to present results for these sectors as they have most projects and make the statistical analysis plausible

15. This pattern is consistent with additional robustness checks in which we re-estimate our main specification while excluding one sector at a time (see Tables B.2 to B.4). In the African sample, the positive and statistically significant birth-region coefficient disappears and moves closer to zero when mining and energy projects are excluded. A similar attenuation occurs when excluding social sector projects, although this may be driven by the substantially smaller sample size, as the coefficient remains close in magnitude to the main estimate. By contrast, the results remain robust when transport projects are excluded.

Figure 4: Effect of Birth Region on Project Placement in PAs by Sector



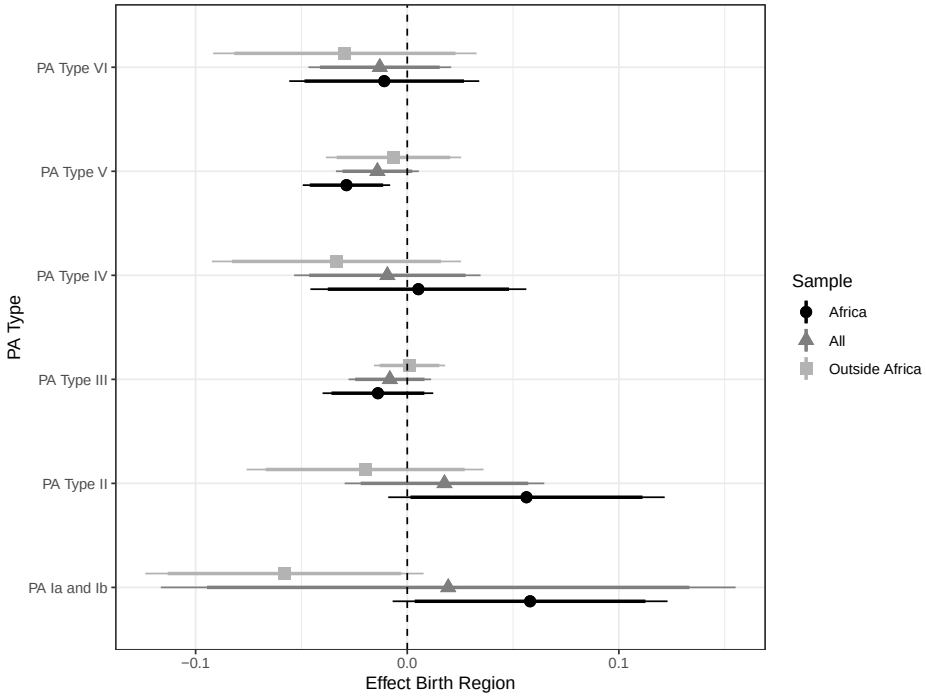
Note: Figure includes point estimates of birth region with 95% and 90% confidence intervals. Standard errors are clustered by recipient country. The models include the following covariates: Number of ADM1 regions covered by project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, and number of PAs. The models also include recipient-year.

resource use. Therefore, we assess whether the birth region dummy is associated with changes in the probability of a project being placed in a PA, depending on the category.

The results in Figure 5 suggest that the birth region effect is most pronounced in the strictest types of PAs. While the findings do not reach conventional significance at the 95% level, likely due to limited statistical power, they are suggestive at the 90% level. In African countries, projects located in birth regions are more likely to be placed in PAs with the highest level of protection, namely IUCN Categories Ia, Ib, and II. In contrast, the opposite pattern emerges in non-African countries.

For less restrictive PAs, we do not observe a birth region effect. In fact, the coefficients turn negative, even for projects in Africa. This finding aligns with our theoretical argument: projects in highly restrictive PAs may offer significant economic benefits but also entail substantial costs, potentially discouraging leaders from pursuing them. In other words, the distinction between projects in birth and non-birth regions is most apparent in situations where violating PA restrictions is particularly costly.

Figure 5: Effect of Birth Region on Project Placement in PAs by Type



Note: Figure includes point estimates of birth region with 95% and 90% confidence intervals. Standard errors are clustered by recipient country. The models include the following covariates: Number of ADM1 regions covered by project, number of mines, number of oil and gas fields, log area, log population in 2000, log average night time light, and number of PAs. The models also include recipient-year and project sector fixed effects. PA types with lower numbers place stricter restrictions on human activites (i.e., PA Types Ia and Ib are the strictest, while Type VI is the least strict).

6 Testing the Scope Conditions

We demonstrated that the placement of aid projects in a political leader's birth region affects the likelihood that these projects are also located within PAs, especially in Africa. In this section, we discuss the role of ethnic voting and public opinion towards China in moderating this relationship.

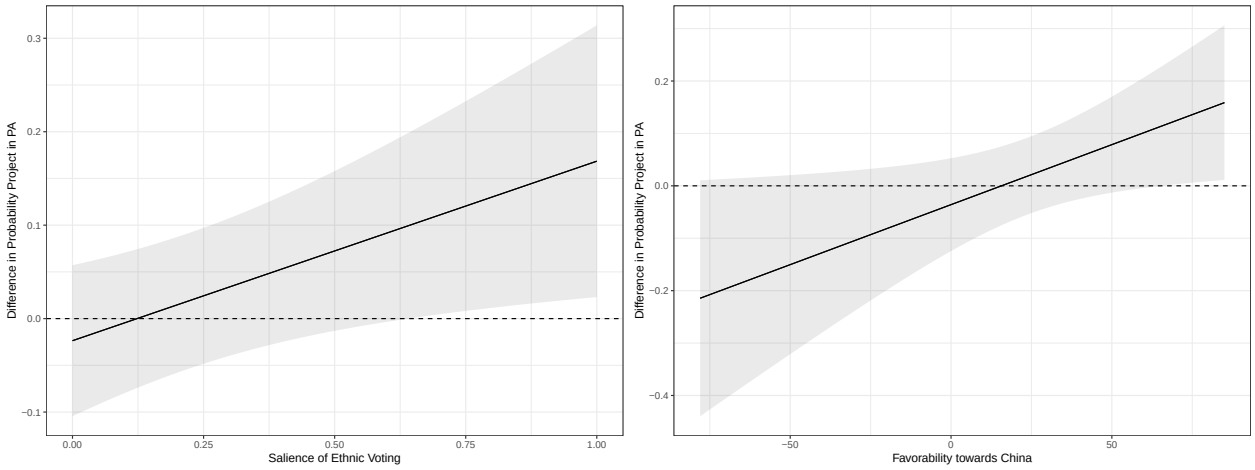
As stated in hypothesis 2, one source of variation in the costs of PA violation is the salience of ethnicity in a polity's politics. We expect that as ethnic salience increases, a project's connection to a leader's home region will become a stronger predictor of its placement within a PA. To evaluate this claim, we interact the home region dummy with the measure of ethnic political voting.

The results in Figure 6a, which are based on the full project sample, provide suggestive evidence in line with our expectation. A project placed in a birth region has no significant relationship with the placement of a project in a PA in countries with low levels of ethnic voting. However, as ethnic voting becomes more salient, we see that the placement of projects in a leader's home region increase the chance of a PA violation. This is in line with our argument that in more ethnically polarized countries, leaders face fewer constraints when dealing with Chinese aid projects.

Figure 6: Moderators of relationship between home region and PA violation

(a) Ethnic political voting

(b) Public opinion towards China



Note: Figure shows marginal effects of birth region on placement in PA conditional the salience of ethnic voting (left panel) and public opinion towards China (right panel) with 95% confidence intervals. Standard errors are clustered by recipient country. The models also include recipient-year and project sector fixed effects.

The second factor that may shape the likelihood of backlash against Chinese projects that infringe on PAs is the favorability of public opinion toward China (see hypothesis 3). We predict that leaders may be more reluctant to place aid projects into a PA when sentiment towards China is not favorable even when it benefits the leader's home region.

Based on the full project sample, we find suggestive evidence that local public opinion can influence how a leader's birth region affects the placement of aid projects in PAs (see Figure 6b). As public opinion toward China becomes more favorable, the relationship between birth region and PA violations reverses. At low levels of favorability, projects located in a leader's birth region are less likely to violate a PA. However, as net favorability increases, projects in a leader's birth region become more likely to be placed in a PA.

Taken together, these findings suggest that violating protected areas constitutes a delicate balancing act between the interests of citizens and those of economic elites. The likelihood that a leader permits encroachment on a PA, particularly to promote development in their home region, increases as the political costs of citizen dissatisfaction decline and the benefits of rewarding coethnic or in-group constituencies rise. This implies that leaders cannot violate protected areas without constraint. Rather, their decisions are shaped by significant political trade-offs and institutional limits.

An obvious question derived from that result is whether the role of good governance and democracy matters as well. It seems plausible that democratic institutions also represent constraint for leaders. We explore this question by interacting the birth-region indicator with a set of V-Dem governance measures (democracy, clientelism, rule of law, and corruption). The interactions are not statistically significant (Table C.3). If anything, better governance is associated with a larger increase in the probability that a project is located in a PA when it benefits a leader's birth region. We cautiously interpret this in line with our previously stated argument that PA violations may serve to obfuscate rent-seeking behavior where formal institutions are stronger, whereas such obfuscation is less necessary in more corrupt or authoritarian contexts.

7 Alternative Explanations

We argue that the placement of aid projects within PAs is largely shaped by the rent-seeking interests of political leaders and the capacity of citizens to hold their leaders accountable. However, there are also other plausible factors that may deter leaders from placing projects in PAs. One such factor is international pressure. Garriga and Zhang (2025) find that countries with larger stocks of foreign direct investment (FDI) tend to have more PAs. They argue that these areas enhance a country's international reputation and help protect incumbent foreign firms by raising entry barriers for potential competitors. As such, foreign firms may have an incentive to pressure

host governments to respect PAs. Similarly, foreign donors may object to environmentally harmful practices or to the violation of domestic environmental laws. Given these dynamics, we might expect that in countries with greater exposure to international scrutiny—measured by higher levels of FDI and a greater share of GDP derived from official development assistance (ODA)—the influence of birth regions on the placement of Chinese aid projects in PAs should be weaker.

To assess these explanations, we interacted the birth region variable with a country's log FDI stock[16] and the share of ODA relative to the gross national income.[17] If the international pressure explanation applies, we should see a negative and statistically significant interaction coefficient. The results in column 1 and 2 of Table 3, however, do not align with that expectation. The coefficients are very close to zero, positive, and statistically insignificant. This suggests that international pressure is unlikely to influence leaders' decisions to place Chinese projects in PAs.

Another possible explanation is that PA violations are a function of development. In countries with lower development, leaders may have a stronger motivation to ignore environmental laws and attract Chinese projects for the sake of promoting growth. To test this, we interacted the birth region indicator with two measures of development: the log-transformed GDP per capita at the country level and the log of nighttime light intensity for ADM1 regions covered by a project. Columns 3 and 4 in Table 3 present the results of this test. If development concerns were driving the pattern, we would expect the interaction terms to be negative, indicating a higher likelihood of PA encroachment in less developed regions. However, we find no statistically significant interaction effects. While the coefficient for the nighttime lights interaction points in the expected direction, it does not reach statistical significance at the 95% level.

Our findings could also be driven by whether a country's political elites benefits from the project. As most of the political elites are located in the national capital of a country, one may hypothesize that projects going to the capital may be particular popular among these elites. Thus, there will be less political opposition when violating a PA. Column 5 in Table 3 indicates projects located in a capital are more likely to be placed inside a PA. This relationship is, however, not statistically significant.

16. Following Garriga and Zhang (2025), we do not look at the annual FDI stock, but we take the rolling average mean across four years.

17. FDI data comes from UNCTAD (2024) and the ODA share data was taken from the World Bank (2024).

Table 3: Testing Alternative Explanations

Dependent Variable:	In PA				
Model:	(i)	(ii)	(iii)	(iv)	(v)
Variables					
Birth Region	0.00329	0.00906	-0.03471	0.03227	-0.03962
	(0.18688)	(0.04627)	(0.29231)	(0.09617)	(0.04347)
Num. ADM1 Regions	0.04022*	0.03976*	0.04058*	0.04023**	0.04012**
	(0.01588)	(0.01542)	(0.01588)	(0.01527)	(0.01511)
Num. Mines	0.00139	0.00233	0.00167	0.00219	0.00204
	(0.00211)	(0.00234)	(0.00210)	(0.00220)	(0.00213)
Num. Oil/Gas	0.01188	0.01036	0.01276*	0.01002	0.00933
	(0.00601)	(0.00578)	(0.00501)	(0.00525)	(0.00532)
Log Population	0.03817*	0.03754*	0.03461*	0.03842*	0.04187*
	(0.01765)	(0.01624)	(0.01595)	(0.01595)	(0.01651)
Log Total Area	0.02761	0.03033	0.03097	0.02975	0.02918
	(0.01666)	(0.01553)	(0.01569)	(0.01532)	(0.01531)
Num. PAs	0.00088*	0.00089*	0.00099**	0.00089*	0.00088*
	(0.00039)	(0.00039)	(0.00037)	(0.00037)	(0.00037)
Log Avg. Night Time Lights	-0.04833	-0.04356	-0.04472	-0.03998	-0.03887
	(0.03469)	(0.03219)	(0.03251)	(0.03150)	(0.03086)
In Capital	0.04183	0.04520	0.04519	0.04506	0.00919
	(0.04019)	(0.03972)	(0.03879)	(0.03845)	(0.03326)
Birth Region × Log FDI Stock	0.00103				
	(0.01965)				
Birth Region × ODA/GNI (in %)		0.00002			
		(0.00369)			
Birth Region × Log GDP/Capita			0.00573		
			(0.03736)		
Birth Region × Log Night Time Ligts				-0.01854	
				(0.04508)	
Birth Region × In Capital					0.11220
					(0.06392)
Fixed-effects					
Recipient-Year	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes
Fit statistics					
R^2	0.574	0.577	0.575	0.580	0.581
Observations	3,430	3,555	3,570	3,675	3,675

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

8 Conclusion

Our findings raise several key implications for scholars of international relations and environmental politics. First, they further temper optimism about the efficacy of PAs. Although there is some evidence that PAs can improve local environmental outcomes (e.g., Miranda et al. 2016; Brodie et al. 2023; Andam et al. 2008), extant work points to political interference that prevents the optimal placement of PAs (Alger 2023; Mangonnet, Kopas, and Urpelainen 2022). Our work points to an additional obstacle to environmental protection: even optimally placed PAs may not be respected if the political incentives for violating them are strong. Nevertheless, PAs create an important focal point for environmental movements to mobilize around. A plausible expectation is that even more aid projects would encroach on environmentally sensitive areas in the absence of PAs.

Second, these findings underscore how political costs can deter leaders from violating PAs. Ethnic political voting and public opinion toward China can influence the costs leaders face when violating PAs, thereby incentivizing them to minimize such violations. These results suggest that stronger accountability mechanisms can help curb elite capture of aid projects that undermine environmental protection (e.g., Böhmelt and Bernauer 2025).

This project highlights several avenues for future research. First, further research is needed to better understand regional differences in the effects of birth regions. Is our finding specific to PA violations, or does it extend to the broader distribution of Chinese aid? Investigating whether birth region favoritism influences aid allocation more generally could provide valuable insights into the political economy of foreign assistance.

Second, survey research on public opinion regarding PA violations could offer new insights into why leaders refrain from violating PAs in some areas while doing so in others. This approach could also help disentangle the historical legacies of PAs and their influence on public attitudes toward conservation.

Third, future research might compare the practices of different donors. While we consider China a most likely case for PA violations and the World Bank a less likely case, we do not investigate why these donors take different approaches. It is possible that bilateral donors prioritize making foreign policy gains and are willing to sacrifice environmental protection. Donors may vary in their interest group environments; perhaps governments in states with large fossil fuel or manufacturing sectors face pressure to fund PA-violating projects, while governments that face large environmentalist interests face pressure to stop such projects.

Fourth, while we analyze the placement of aid projects in PAs, an important next step is to

assess their environmental consequences. We assume that placing infrastructure projects, such as energy plants, within PAs has significant environmental impacts, but this remains an open empirical question. Future research should investigate the extent of these impacts, assessing whether aid projects within PAs lead to measurable environmental degradation or if certain mitigation strategies are effective in minimizing harm.

Our work demonstrates that international development efforts can interact with domestic political incentives to undermine environmental protection efforts. Crucially, this and other work in environmental politics points to a dire need to take domestic politics seriously in the implementation of environmental protection efforts. If, for example, recent efforts to mobilize billions of dollars in green finance are ignorant of the threat of political interference, they risk falling short of their important goals.

References

Alesina, Alberto, and Beatrice Weder. 2002. "Do Corrupt Governments Receive Less Foreign Aid?" *American Economic Review* 92 (4): 1126–1137. https://doi.org/10.1257/000282802603446669.

Alger, Justin. 2023. "The Political Economy of Protected Area Designations: Commercial Interests in Conservation Policy." *Global Environmental Politics* 23 (2): 54–73. https://doi.org/10.1162/glep_a_00690.

Anaxagorou, Christiana, Georgios Efthyvoulou, and Vassilis Sarantides. 2020. "Electoral motives and the subnational allocation of foreign aid in sub-Saharan Africa." *European Economic Review* 127:103430. https://doi.org/10.1016/j.euroecorev.2020.103430.

Andam, Kwaw S., Paul J. Ferraro, Alexander Pfaff, G. Arturo Sanchez-Azofeifa, and Juan A. Robalino. 2008. "Measuring the effectiveness of protected area networks in reducing deforestation." *Proceedings of the National Academy of Sciences* 105 (42): 16089–16094. https://doi.org/10.1073/pnas.0800437105.

Andersen, Jørgen Juel, Niels Johannesen, and Bob Rijkers. 2022. "Elite Capture of Foreign Aid: Evidence from Offshore Bank Accounts." *Journal of Political Economy* 130 (2): 388–425. https://doi.org/10.1086/717455.

André, Pierre, Paul Maarek, and Fatoumata Tapo. 2025. "Can donors prevent aid misallocations? Evidence from Chinese and World Bank aid." *World Development* 186:106811. https://doi.org/10.1016/j.worlddev.2024.106811.

Baehr, Christian, Ariel BenYishay, and Bradley C. Parks. 2021. "Linking Local Infrastructure Development and Deforestation: Evidence from Satellite and Administrative Data." *Journal of the Association of Environmental and Resource Economists* 8 (2): 375–409. https://doi.org/10.1086/712800.

———. 2023. "Highway to the forest? Land governance and the siting and environmental impacts of Chinese government-funded road building in Cambodia." *Journal of Environmental Economics and Management* 122:102898. https://doi.org/10.1016/j.jeem.2023.102898.

Baldwin, Kate, and Matthew S. Winters. 2023. "Foreign Aid and Political Support: How Politicians' Aid Oversight Capacity and Voter Information Condition Credit-Giving." *World Politics* 75 (1): 1–42. https://doi.org/10.1353/wp.2023.0004..

Beacham, Austin. 2023. *Extraction, Contestation, and Conservation: Natural Resource Dependence and Protected Area Designation.* Technical report 22. San Diego: IGCC. Accessed September 5, 2024. https://ucigcc.org/wp-content/uploads/2023/11/2023_wp22_beacham-v2-FINAL.pdf.

Beesley, Celeste, and Alexander Slaski. 2024. "Jobs first, environment second: the conditional effect of pollution on perceptions of Foreign Direct Investment." *Review of International Political Economy* 31 (6): 1662–1685. https://doi.org/10.1080/09692290.2024.2367576.

BenYishay, Ariel, Bradley Parks, Daniel Runfola, and Rachel Trichler. 2016. *Forest Cover Impacts of Chinese Development Projects in Ecologically Sensitive Areas.* Working Paper 32. Williamsburg: AidData.

Berlin, Maria Perrotta, Raj M. Desai, and Anders Olofsgård. 2023. "Trading favors? UN Security Council membership and subnational favoritism in aid recipients." *Review of International Organizations* 18 (2): 237–258. https://doi.org/10.1007/s11558-022-09467-z.

Bjørnskov, Christian. 2010. "Do elites benefit from democracy and foreign aid in developing countries?" *Journal of Development Economics* 92 (2): 115–124. https://doi.org/10.1016/j.jdeveco.2009.03.001.

Böhmelt, Tobias, and Thomas Bernauer. 2025. "New evidence reveals curvilinear relationship between levels of democracy and deforestation." *Research & Politics* 12 (1): 20531680251320073. https://doi.org/10.1177/20531680251320073.

Bomprezzi, Pietro, Axel Dreher, Andreas Fuchs, Teresa Hailer, Andreas Kammerlander, Lennart Kaplan, Silvia Marchesi, Tania Masi, Charlotte Robert, and Kerstin Unfried. 2024a. *The Political Leaders' Affiliation Database (PLAD).* https://doi.org/10.7910/DVN/YUS575. https://dataverse.harvard.edu/dataset.xhtml?persistentId=doi:10.7910/DVN/YUS575.

———. 2024b. *Wedded to Prosperity? Informal Influence and Regional Favoritism.* CESifo Working Paper 10969. Munich: CESifo. Accessed September 15, 2024.

Bontempi, Antonio, Pietro Venturi, Daniela Del Bene, Arnim Scheidel, Quim Zaldo-Aubanell, and Roser Maneja Zaragoza. 2023. "Conflict and conservation: On the role of protected areas for environmental justice." *Global Environmental Change* 82:102740. https://doi.org/10.1016/j.gloenvcha.2023.102740.

Brazys, Samuel, Johan A. Elkink, and Gina Kelly. 2017. "Bad neighbors? How co-located Chinese and World Bank development projects impact local corruption in Tanzania." *The Review of International Organizations* 12 (2): 227–253. https://doi.org/10.1007/s11558-017-9273-4.

Brazys, Samuel, and Krishna Chaitanya Vadlamannati. 2021. "Aid curse with Chinese characteristics? Chinese development flows and economic reforms." *Public Choice* 188 (3): 407–430. https://doi.org/10.1007/s11127-020-00836-z.

Briggs, Ryan C. 2021. "Why Does Aid Not Target the Poorest?" *International Studies Quarterly* 65 (3): 739–752. https://doi.org/10.1093/isq/sqab035.

Brodie, Jedediah F., Jayasilan Mohd-Azlan, Cheng Chen, Oliver R. Wearn, Mairin C. M. Deith, James G. C. Ball, Eleanor M. Slade, et al. 2023. "Landscape-scale benefits of protected areas for tropical biodiversity." *Nature* 620 (7975): 807–812. https://doi.org/10.1038/s41586-023-06410-z.

Brown, Kathleen J. 2025. "Why hide? Africa's unreported debt to China." *The Review of International Organizations* 20 (1): 1–32. https://doi.org/10.1007/s11558-023-09513-4.

Buntaine, Mark T. 2015. "Accountability in Global Governance: Civil Society Claims for Environmental Performance at the World Bank." *International Studies Quarterly* 59 (1): 99–111. https://doi.org/10.1111/isqu.12145.

Burnside, Craig, and David Dollar. 2000. "Aid, Policies, and Growth." *American Economic Review* 90 (4): 847–868. https://doi.org/10.1257/aer.90.4.847.

Chen, Alicia R. 2025. "Electoral Incentives and the Choice of Infrastructure Development Aid." Stanford University. Accessed December 1, 2025. https://www.dropbox.com/scl/fi/tn0u62l14olowl91hh1iw/arc_jmp_speed.pdf?rlkey=7lbwpd3phinnj6pd5qufeq4yn&e=1&dl=0.

Chubb, Andrew. 2024. *Global Public Opinion on China 1.0 dataset.* Accessed March 3, 2025. https://asiasociety.org/policy-institute/global-public-opinion-china.

Cormier, Ben. 2023. "Chinese or western finance? Transparency, official credit flows, and the international political economy of development." *The Review of International Organizations* 18 (2): 297–328. https://doi.org/10.1007/s11558-022-09469-x.

Cruz, Cesi, and Christina J. Schneider. 2017. "Foreign Aid and Undeserved Credit Claiming." *American Journal of Political Science* 61 (2): 396–408. https://doi.org/10.1111/ajps.12285.

Dawson, Stephen, Felix Haass, Carl Müller-Crepon, and Aksel Sundström. 2025. "The Ethnic Politics of Nature Protection in Africa." *The Journal of Politics,* 739777. https://doi.org/10.1086/739777.

Dietrich, Simone, Minhaj Mahmud, and Matthew S. Winters. 2018. "Foreign Aid, Foreign Policy, and Domestic Government Legitimacy: Experimental Evidence from Bangladesh." *Journal of Politics* 80 (1): 133–148. https://doi.org/10.1086/694235.

Dreher, Axel, Vera Z Eichenauer, and Kai Gehring. 2018. "Geopolitics, Aid, and Growth: The Impact of UN Security Council Membership on the Effectiveness of Aid." *The World Bank Economic Review* 32 (2): 268–286. https://doi.org/10.1093/wber/lhw037.

Dreher, Axel, Andreas Fuchs, Roland Hodler, Bradley C. Parks, Paul A. Raschky, and Michael J. Tierney. 2019. "African leaders and the geography of China's foreign assistance." *Journal of Development Economics* 140:44–71. https://doi.org/10.1016/j.jdeveco.2019.04.003.

———. 2021. "Is Favoritism a Threat to Chinese Aid Effectiveness? A Subnational Analysis of Chinese Development Projects." *World Development* 139:105291. https://doi.org/10.1016/j.worlddev.2020.105291.

Dreher, Axel, Andreas Fuchs, Bradley Parks, Austin Strange, and Michael J. Tierney. 2021. "Aid, China, and Growth: Evidence from a New Global Development Finance Dataset." *American Economic Journal: Economic Policy* 13 (2): 135–174.

———. 2022. *Banking on Beijing: The Aims and Impacts of China's Overseas Development Program.* Cambridge: Cambridge University Press. ISBN: 978-1-108-47410-8.

EJ Atlas. 2022a. *Mattala Airport, Sri Lanka.*

———. 2022b. *White River, Molleturo, Azuay, Ecuador.*

———. 2023. *East Africa Crude Oil Pipeline, Uganda and Tanzania.* Accessed August 19, 2024. https://ejatlas.org/print/east-africa-crude-oil-pipeline.

Garriga, Ana Carolina, and Muzhou Zhang. 2025. "Are Foreign Firms Good for the Environment? FDI and Protected Areas." *International Interactions* 51 (2): 224–263. https://doi.org/10.1080/03050629.2025.2473362.

Global Witness. 2021. *Major dam project in one of Africa's oldest national parks could breach Congolese law and threatens forests and biodiversity in DRC, with Chinese dam-builder backed by international investors.* Accessed February 20, 2026. https://globalwitness.org/en/press-releases/major-dam-project-in-one-of-africas-oldest-national-parks-could-breach-congolese-law-and-threatens-forests-and-biodiversity-in-drc-with-chinese-dam-builder-backed-by-international-investors/.

Goodman, Seth, Sheng Zhang, Ammar A. Malik, Bradley C. Parks, and Jacob Hall. 2024. "AidData's Geospatial Global Chinese Development Finance Dataset." *Scientific Data* 11 (1): 529. https://doi.org/10.1038/s41597-024-03341-w.

Hodler, Roland. 2007. "Rent seeking and aid effectiveness." *International Tax and Public Finance* 14 (5): 525–541. https://doi.org/10.1007/s10797-006-9006-8.

Hodler, Roland, and Paul A. Raschky. 2014. "Regional Favoritism." *Quarterly Journal of Economics* 129 (2): 995–1033. https://doi.org/10.1093/qje/qju004.

Holland, Alisha C. 2015. "The Distributive Politics of Enforcement." *American Journal of Political Science* 59 (2): 357–371. https://doi.org/10.1111/ajps.12125.

Honaker, James, Gary King, and Matthew Blackwell. 2011. "Amelia II: A Program for Missing Data." *Journal of Statistical Software* 45 (7). https://doi.org/10.18637/jss.v045.i07.

Isaksson, Ann-Sofie, and Andreas Kotsadam. 2018a. "Chinese aid and local corruption." *Journal of Public Economics* 159:146–159. https://doi.org/10.1016/j.jpubeco.2018.01.002.

———. 2018b. "Racing to the bottom? Chinese development projects and trade union involvement in Africa." *World Development* 106:284–298. https://doi.org/10.1016/j.worlddev.2018.02.003.

Jablonski, Ryan S. 2014. "How Aid Targets Votes: The Impact of Electoral Incentives on Foreign Aid Distribution." *World Politics* 66 (2): 293–330. https://doi.org/10.1017/S0043887114000045.

Juon, Andreas. 2024. "The long-term consequences of power-sharing for ethnic salience." *Journal of Peace Research,* https://doi.org/10.1177/00223433241227759.

Kaplan, Stephen B. 2021. *Globalizing Patient Capital: The Political Economy of Chinese Finance in the Americas.* Cambridge: Cambridge University Press. ISBN: 978-1-107-18231-8.

Kasara, Kimuli. 2007. "Tax Me If You Can: Ethnic Geography, Democracy, and the Taxation of Agriculture in Africa." *American Political Science Review* 101 (1): 159–172.

Kersting, Erasmus K., and Christopher Kilby. 2016. "With a little help from my friends: Global electioneering and World Bank lending." *Journal of Development Economics* 121:153–165. https://doi.org/10.1016/j.jdeveco.2016.03.010.

Knox, Dean, Will Lowe, and Jonathan Mummolo. 2020. "Administrative Records Mask Racially Biased Policing." *American Political Science Review* 114 (3): 619–637. https://doi.org/10.1017/S0003055420000039.

Labonne, Julien. 2016. "Local political business cycles: Evidence from Philippine municipalities." *Journal of Development Economics* 121:56–62. https://doi.org/10.1016/j.jdeveco.2016.03.004.

Li, Xiaojun. 2017. "Does Conditionality Still Work? China's Development Assistance and Democracy in Africa." *Chinese Political Science Review* 2 (2): 201–220. https://doi.org/10.1007/s41111-017-0050-6.

Licht, Amanda A. 2010. "Coming into Money: The Impact of Foreign Aid on Leader Survival." *Journal of Conflict Resolution* 54 (1): 58–87. https://doi.org/10.1177/0022002709351104.

Mangonnet, Jorge, Jacob Kopas, and Johannes Urpelainen. 2022. "Playing Politics with Environmental Protection: The Political Economy of Designating Protected Areas." *Journal of Politics* 84 (3): 1453–1468. https://doi.org/10.1086/718978.

Marchesi, Silvia, Tania Masi, and Saumik Paul. 2025. "Aid Projects and Firm Performance." *Economic Development and Cultural Change* 73 (3): 1251–1295. https://doi.org/10.1086/730829.

Marty, Robert, Seth Goodman, Michael LeFew, Carrie Dolan, Ariel BenYishay, and Daniel Runfola. 2019. "Assessing the causal impact of Chinese aid on vegetative land cover in Burundi and Rwanda under conditions of spatial imprecision." *Development Engineering* 4:100038. https://doi.org/10.1016/j.deveng.2018.11.001.

Masaki, Takaaki. 2018. "The Political Economy of Aid Allocation in Africa: Evidence from Zambia." *African Studies Review* 61 (1): 55–82. https://doi.org/10.1017/asr.2017.97.

Mattingly, Daniel C. 2016. "Elite Capture: How Decentralization and Informal Institutions Weaken Property Rights in China." *World Politics* 68 (3): 383–412. https://doi.org/10.1017/S0043887116000083.

Michaelsen, Patrik, Aksel Sundström, and Sverker C. Jagers. 2025. "Mass support for conserving 30% of the Earth by 2030: Experimental evidence from five continents." *Proceedings of the National Academy of Sciences* 122 (35): e2503355122. https://doi.org/10.1073/pnas.2503355122.

Miller, Tom. 2019. *China's Asian Dream: Empire Building along the New Silk Road.* 2nd edition. London: Zed Books. ISBN: 978-1-78699-741-8.

Miranda, Juan José, Leonardo Corral, Allen Blackman, Gregory Asner, and Eirivelthon Lima. 2016. "Effects of Protected Areas on Forest Cover Change and Local Communities: Evidence from the Peruvian Amazon." *World Development* 78:288–307. https://doi.org/10.1016/j.worlddev.2015.10.026.

Moehlecke, Carolina, Guilherme N. Fasolin, and Matias Spektor. 2025. "Beyond Jobs: When Citizens Reject Socially Irresponsible Foreign Direct Investment." *International Studies Quarterly* 69 (3): sqaf046. https://doi.org/10.1093/isq/sqaf046.

Naughton-Treves, Lisa, Margaret Buck Holland, and Katrina Brandon. 2005. "The Role of Protected Areas in Conserving Biodiversity and Sustaining Local Livelihoods." *Annual Review of Environment and Resources* 30:219–252. https://doi.org/10.1146/annurev.energy.30.050504.164507.

Newmark, William D., Nancy L. Leonard, Hashim I. Sariko, and Deo-Gratias M. Gamassa. 1993. "Conservation attitudes of local people living adjacent to five protected areas in Tanzania." *Biological Conservation* 63 (2): 177–183. https://doi.org/10.1016/0006-3207(93)90507-W.

Nguyen, Quynh, and Edmund Malesky. 2021. "Fish or steel? New evidence on the environment-economy trade-off in developing Vietnam." *World Development* 147:105603. ISSN: 0305-750X. https://doi.org/10.1016/j.worlddev.2021.105603.

Shea, Patrick E., Bernhard Reinsberg, and Andreas Kern. 2025. "China Lending and the Political Economy of Leader Survival." *Journal of Conflict Resolution* 69 (4): 571–598. https://doi.org/10.1177/00220027241263036.

Springman, Jeremy. 2022. "The Political Economy of NGO Service Provision: Evidence from an Ancillary Field Experiment in Uganda." *World Politics* 74 (4): 523–563. https://doi.org/10.1017/S0043887122000107.

Strange, Austin. 2024. "Symbols of State: Explaining Prestige Projects in the Global South." *International Studies Quarterly* 68 (2): sqae049. https://doi.org/10.1093/isq/sqae049.

Sun, Xin. 2021. "Politicised enforcement in China: evidence from the enforcement of land laws and regulations." *Journal of Public Policy* 41 (1): 66–89. https://doi.org/10.1017/S0143814X19000321.

Sundström, Aksel. 2016. "Corruption and Violations of Conservation Rules: A Survey Experiment with Resource Users." *World Development* 85:73–83. https://doi.org/10.1016/j.worlddev.2016.04.011.

Svensson, Jakob. 2000. "Foreign aid and rent-seeking." *Journal of International Economics* 51 (2): 437–461. https://doi.org/10.1016/S0022-1996(99)00014-8.

UNCTAD. 2024. *Foreign direct investment: Inward and outward flows and stock, annual.* Https://unctadstat.unctad.org/datacentre/dataviewer/US.FdiFlowsStock (last accessed May 14, 2025).

UNEP-WCMC. 2019. *User Manual for the World Database on Protected Areas and world database on other effective area-based conservation measures.* Cambridge, UK. Accessed September 6, 2024. https://wdpa.s3-eu-west-1.amazonaws.com/WDPA_Manual/English/WDPA_WDOECM_Manual_1_6.pdf.

Wang, Yuan. 2022. "Presidential extraversion: Understanding the politics of Sino-African mega-infrastructure projects." *World Development* 158:105976. ISSN: 0305-750X. https://doi.org/10.1016/j.worlddev.2022.105976.

Welz, Adam. 2019. *How Kenya's Push for Development Is Threatening Its Famed Wild Lands.* Https://e360.yale.edu/features/how-kenyas-push-for-development-is-threatening-its-prized-wild-lands (last accessed July 24, 2025).

Williams, Martin J. 2017. "The Political Economy of Unfinished Development Projects: Corruption, Clientelism, or Collective Choice?" *American Political Science Review* 111 (4): 705–723. https://doi.org/10.1017/S0003055417000351.

World Bank. 2024. *World Bank Open Data.* Https://data.worldbank.org/ (last accessed May 14, 2025). https://data.worldbank.org/.

Xiong, Guoling, Fan Yang, Tongli Wang, Rongxiao He, and Lanxi Li. 2025. "Impact of road infrastructure on wildlife corridors in Hainan rainforests." *Transportation Research Part D: Transport and Environment* 139:104539. https://doi.org/10.1016/j.trd.2024.104539.

Yuichi Kono, Daniel, and Gabriella R. Montinola. 2009. "Does Foreign Aid Support Autocrats, Democrats, or Both?" *The Journal of Politics* 71 (2): 704–718. https://doi.org/10.1017/S0022381609090550.

Zeitz, Alexandra O. 2021. "Emulate or differentiate? Chinese development finance, competition, and World Bank infrastructure funding." *Review of International Organizations* 16 (2): 265–292. https://doi.org/10.1007/s11558-020-09377-y.

Supplementary Material

A Full Main Results

Table A.1: Effect of Birth Region on PA Placement

Dependent Variable:	In PA							
Model:	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Variables								
Birth Region	0.170**	0.044	0.039	0.031	0.011	0.010	0.010	0.006
	(0.034)	(0.040)	(0.040)	(0.039)	(0.046)	(0.044)	(0.044)	(0.041)
Num. ADM1 Regions		0.070**	0.067**	0.063**	0.046**	0.041**	0.041**	0.041**
		(0.014)	(0.014)	(0.015)	(0.014)	(0.015)	(0.015)	(0.015)
Num. Mines			0.007*	0.007*	0.003	0.002	0.002	0.002
			(0.003)	(0.003)	(0.002)	(0.002)	(0.002)	(0.002)
Num. Oil/Gas				0.014*	0.009	0.010*	0.010*	0.010
				(0.006)	(0.005)	(0.005)	(0.005)	(0.005)
Log Population					0.040*	0.037*	0.043*	0.036*
					(0.016)	(0.016)	(0.017)	(0.016)
Log Total Area					0.043**	0.039**	0.030*	0.034*
					(0.009)	(0.008)	(0.014)	(0.015)
Num. PAs						0.001*	0.001*	0.001*
						(0.000)	(0.000)	(0.000)
Log Avg. Nighttime Light							-0.025	-0.032
							(0.030)	(0.032)
In Capital								0.043
								(0.038)
Fixed-effects								
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
R^2	0.470	0.558	0.561	0.563	0.579	0.582	0.582	0.583
Observations	3,875	3,875	3,875	3,875	3,875	3,875	3,875	3,875

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Table A.2: Effect of Birth Region on PA Placement in Africa

Dependent Variable:	In PA							
Model:	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Variables								
Birth Region	0.214**	0.096*	0.095*	0.094*	0.096*	0.094*	0.101*	0.099*
	(0.038)	(0.038)	(0.039)	(0.036)	(0.043)	(0.042)	(0.041)	(0.039)
Num. ADM1 Regions		0.099**	0.096**	0.096**	0.074**	0.068**	0.072**	0.072**
		(0.010)	(0.011)	(0.010)	(0.010)	(0.013)	(0.014)	(0.014)
Num. Mines			0.002	0.002	0.000	0.000	0.000	0.000
			(0.002)	(0.002)	(0.003)	(0.003)	(0.003)	(0.003)
Num. Oil/Gas				0.000	0.008	0.010	0.008	0.008
				(0.017)	(0.020)	(0.018)	(0.018)	(0.017)
Log Population					0.030	0.030	0.045*	0.044**
					(0.017)	(0.018)	(0.018)	(0.016)
Log Total Area					0.036**	0.035**	-0.001	0.000
					(0.009)	(0.009)	(0.019)	(0.020)
Num. PAs						0.000	0.000	0.000
						(0.000)	(0.000)	(0.000)
Log Avg. Night Time Light							-0.098*	-0.099*
							(0.044)	(0.045)
In Capital								0.007
								(0.047)
Fixed-effects								
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
R^2	0.457	0.591	0.591	0.591	0.603	0.604	0.607	0.607
Observations	1,850	1,850	1,850	1,850	1,850	1,850	1,850	1,850

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Table A.3: Effect of Birth Region on PA Placement Outside of Africa

Dependent Variable:	In PA							
Model:	(i)	(ii)	(iii)	(iv)	(v)	(vi)	(vii)	(viii)
Variables								
Birth Region	0.103*	-0.030	-0.039	-0.057	-0.112*	-0.116*	-0.113*	-0.112*
	(0.048)	(0.059)	(0.059)	(0.056)	(0.054)	(0.053)	(0.052)	(0.052)
Num. ADM1 Regions		0.055**	0.053**	0.047**	0.035*	0.032*	0.032*	0.031*
		(0.016)	(0.016)	(0.015)	(0.014)	(0.015)	(0.014)	(0.014)
Num. Mines			0.008	0.006	0.003	0.003	0.003	0.003
			(0.005)	(0.004)	(0.003)	(0.003)	(0.003)	(0.003)
Num. Oil/Gas				0.022**	0.014*	0.014*	0.014*	0.013*
				(0.007)	(0.006)	(0.006)	(0.006)	(0.006)
Log Population					0.049*	0.045*	0.037	0.032
					(0.020)	(0.022)	(0.027)	(0.027)
Log Total Area					0.043**	0.040*	0.050*	0.052*
					(0.016)	(0.015)	(0.024)	(0.024)
Num. PAs						0.001	0.001	0.001
						(0.001)	(0.001)	(0.001)
Log Avg. Night Time Light							0.025	0.020
							(0.044)	(0.044)
In Capital								0.034
								(0.038)
Fixed-effects								
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics								
R^2	0.498	0.562	0.565	0.574	0.588	0.589	0.589	0.590
Observations	2,025	2,025	2,025	2,025	2,025	2,025	2,025	2,025

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

B Robustness Tests

Table B.1: Effect of Birth Region on Placement in PA with Buffer

	All		Africa		No Africa	
Model:	In PA (1km) (i)	In PA (5km) (ii)	In PA (1km) (iii)	In PA (5km) (iv)	In PA (1km) (v)	In PA (5km) (vi)
Variables						
Birth Region	0.022	0.012	0.109**	0.062	-0.082	-0.078
	(0.038)	(0.042)	(0.040)	(0.046)	(0.050)	(0.068)
Num. ADM1 Regions	0.041*	0.027*	0.080**	0.053**	0.028+	0.017
	(0.016)	(0.013)	(0.013)	(0.014)	(0.015)	(0.013)
Num. Mines	0.005*	0.009**	-0.001	0.002	0.010**	0.016**
	(0.002)	(0.003)	(0.003)	(0.003)	(0.002)	(0.005)
Num. Oil/Gas	0.008	0.005	0.000	0.013	0.012+	0.002
	(0.006)	(0.008)	(0.016)	(0.018)	(0.007)	(0.009)
Log Population	0.055**	0.102**	0.065**	0.112**	0.053+	0.097**
	(0.019)	(0.024)	(0.018)	(0.028)	(0.030)	(0.036)
Log Total Area	0.000	-0.001	-0.029	-0.024	0.015	0.033
	(0.017)	(0.023)	(0.023)	(0.028)	(0.028)	(0.040)
Num. PAs	0.001*	0.000	0.000	0.000	0.001	0.000
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
Log Avg. Night Time Light	-0.078+	-0.047	-0.116+	-0.066	-0.047	-0.005
	(0.040)	(0.055)	(0.067)	(0.087)	(0.042)	(0.066)
In Capital	0.037	0.040	-0.007	0.020	0.015	0.006
	(0.043)	(0.046)	(0.065)	(0.071)	(0.044)	(0.048)
Fixed-effects						
Recipient-Year	Yes	Yes	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes	Yes	Yes
Fit statistics						
R^2	0.583	0.605	0.607	0.626	0.585	0.607
Observations	3,675	3,675	1,850	1,850	1,825	1,825

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05, +: 0.1*

Table B.2: Main Results Excluding Transport Projects

Dependent Variable:	In PA		
	All	Africa	No Africa
Model:	(i)	(ii)	(iii)
Variables			
Birth Region	0.044	0.118**	-0.089
	(0.039)	(0.037)	(0.057)
Num. ADM1 Regions	0.041*	0.073**	0.028
	(0.018)	(0.014)	(0.018)
Num. Mines	0.001	0.000	0.002
	(0.002)	(0.003)	(0.003)
Num. Oil/Gas	0.007	0.001	0.010
	(0.005)	(0.015)	(0.006)
Log Population	0.038	0.028	0.053
	(0.022)	(0.023)	(0.036)
Log Total Area	0.034	0.007	0.047
	(0.021)	(0.027)	(0.035)
Num. PAs	0.001	0.000	0.001
	(0.000)	(0.000)	(0.001)
Log Avg. Nighttime Light	-0.043	-0.096	-0.004
	(0.038)	(0.052)	(0.054)
In Capital	0.032	0.030	-0.015
	(0.045)	(0.053)	(0.052)
Fixed-effects			
Recipient-Year	Yes	Yes	Yes
Sector	Yes	Yes	Yes
Fit statistics			
R^2	0.625	0.653	0.628
Observations	2,933	1,493	1,440

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Table B.3: Main Results Excluding Mining and Energy Projects

Dependent Variable:	In PA		
	All	Africa	No Africa
Model:	(i)	(ii)	(iii)
Variables			
Birth Region	-0.008	0.069	-0.136*
	(0.042)	(0.043)	(0.060)
Num. ADM1 Regions	0.045*	0.089**	0.022
	(0.021)	(0.012)	(0.018)
Num. Mines	0.005	0.000	0.018*
	(0.003)	(0.003)	(0.007)
Num. Oil/Gas	0.007	0.021	0.007
	(0.009)	(0.024)	(0.008)
Log Population	0.033	0.045**	0.016
	(0.017)	(0.016)	(0.035)
Log Total Area	0.030	-0.012	0.069*
	(0.021)	(0.024)	(0.028)
Num. PAs	0.001	0.000	0.001
	(0.000)	(0.000)	(0.001)
Log Avg. Nighttime Light	-0.056	-0.102	0.010
	(0.043)	(0.059)	(0.063)
In Capital	0.081	-0.008	0.116*
	(0.048)	(0.050)	(0.056)
Fixed-effects			
Recipient-Year	Yes	Yes	Yes
Sector	Yes	Yes	Yes
Fit statistics			
R^2	0.640	0.651	0.670
Observations	2,549	1,509	1,040

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

Table B.4: Main Results Excluding Social Projects

Dependent Variable:	In PA		
	All	Africa	No Africa
Model:	(i)	(ii)	(iii)
Variables			
Birth Region	-0.025	0.086	-0.123*
	(0.051)	(0.068)	(0.055)
Num. ADM1 Regions	0.030	0.057**	0.028
	(0.016)	(0.016)	(0.017)
Num. Mines	0.000	-0.002	0.000
	(0.003)	(0.004)	(0.003)
Num. Oil/Gas	0.014*	0.008	0.017*
	(0.005)	(0.012)	(0.007)
Log Population	0.039*	0.041*	0.042
	(0.019)	(0.020)	(0.030)
Log Total Area	0.041*	0.026	0.043
	(0.017)	(0.021)	(0.034)
Num. PAs	0.001**	0.001	0.001
	(0.000)	(0.001)	(0.001)
Log Avg. Nighttime Light	-0.029	-0.077	0.001
	(0.039)	(0.054)	(0.059)
In Capital	0.057	0.034	0.024
	(0.047)	(0.070)	(0.047)
Fixed-effects			
Recipient-Year	Yes	Yes	Yes
Sector	Yes	Yes	Yes
Fit statistics			
R^2	0.631	0.661	0.624
Observations	2,447	1,044	1,403

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

C Other Plausible Explanations

C.1 Birth Region to Non Birth Region Differences

Tables C.1 and C.2 document systematic differences between birth regions and non-birth regions. We compare basic geographic and economic characteristics of birth regions and non birth regions in two samples. In Table C.1, we look at how the characteristics differ between projects that benefit a birth region and projects that do not benefit any birth region. Table C.2 compares the differences across all geographic ADM1 regions independent of the project.

Table C.1: Balance table: Comparing Birth Region Projcts to Non-Birth Region Projects

Variable	Mean Birth Region	Mean Non Birth Region	Difference-in-Means	t Statistic	$p < 0.05$
Number of PAs	27.90	16.98	10.93	-4.59	Yes
Capital	0.60	0.35	0.25	-13.21	Yes
Log Area	9.82	9.44	0.38	-4.45	Yes
Log Night Time Lights	1.74	1.75	-0.00	0.20	No
Log Population	5.19	4.91	0.29	-4.34	Yes
Log Population Density	14.99	14.31	0.68	-10.61	Yes
Number of Mines	3.01	1.87	1.14	-3.58	Yes
Number Oil Fields	1.70	0.83	0.88	-6.71	Yes

Table C.2: Balance table: Comparing ADM1 Birth Regions to Non-Birth Regions

Variable	Mean Birth Region	Mean Non Birth Region	Difference-in-Means	t Statistic	$p < 0.05$
Number of PAs	11.65	10.31	1.34	-0.88	No
Capital	0.20	0.03	0.17	-7.60	Yes
Log Area	9.00	8.54	0.47	-3.71	Yes
Log Night Time Lights	1.55	1.32	0.23	-4.46	Yes
Log Population	13.77	12.78	0.99	-9.62	Yes
Log Population Density	4.77	4.25	0.52	-4.50	Yes
Number of Mines	1.20	0.78	0.42	-1.60	No
Number Oil Fields	0.45	0.53	-0.08	1.32	No

C.2 Governance quality

A common concern is whether our main results vary with widely used governance indicators. In Table C.3, we interact the birth–region indicator with four V-Dem measures—democracy, corruption, clientelism, and rule of law. Across all specifications, none of these variables significantly moderates the birth–region effect: the interaction terms are uniformly statistically indistinguishable from zero.

Table C.3: Effect of Birth Region and Governance

Dependent Variable:	In PA			
Model:	(i)	(ii)	(iii)	(iv)
Variables				
Birth Region	-0.063	0.155	0.155	-0.059
	(0.100)	(0.094)	(0.120)	(0.080)
Num. ADM1 Regions	0.041**	0.041**	0.041**	0.041**
	(0.016)	(0.015)	(0.015)	(0.016)
Num. Mines	0.002	0.002	0.002	0.002
	(0.002)	(0.002)	(0.002)	(0.002)
Num. Oil/Gas	0.011*	0.011*	0.011*	0.011*
	(0.005)	(0.005)	(0.005)	(0.005)
Log Population	0.038*	0.039*	0.038*	0.038*
	(0.016)	(0.016)	(0.016)	(0.016)
Log Total Area	0.028	0.028	0.028	0.027
	(0.015)	(0.015)	(0.015)	(0.015)
Num. PAs	0.001*	0.001*	0.001*	0.001*
	(0.000)	(0.000)	(0.000)	(0.000)
Log Avg. Night Time Light	-0.048	-0.046	-0.046	-0.049
	(0.031)	(0.031)	(0.031)	(0.030)
In Capital	0.048	0.047	0.046	0.048
	(0.039)	(0.040)	(0.040)	(0.040)
Birth Region × Electoral Democracy	0.183			
	(0.187)			
Birth Region × Corruption Index		-0.212		
		(0.147)		
Birth Region × Clientelism Index			-0.245	
			(0.194)	
Birth Region × Rule of Law				0.195
				(0.153)
Fixed-effects				
Recipient-Year	Yes	Yes	Yes	Yes
Sector	Yes	Yes	Yes	Yes
Fit statistics				
R^2	0.580	0.581	0.581	0.581
Observations	3,660	3,660	3,660	3,660

Clustered (Recipient) standard-errors in parentheses
*Signif. Codes: **: 0.01, *: 0.05*

D Adjusting for Post-Treatment Bias

Assumptions

Studying the effect of race on use of force in police encounters, Knox, Lowe, and Mummolo (2020) describe a procedure for estimation of the average treatment effect while conditioning on a post-treatment moderator (in their case, whether an encounter occurred at all). We adapt this procedure to our study, allowing us to bound the ATE of a leader's birth region on the placement of a Chinese project in a PA (PA), conditional on a project being placed (which is itself a function of a leader's birth region). We formally describe the procedure in our context below.

Let the outcome Y_i be a dummy variable that takes value 1 if a project i is placed in a PA and 0 otherwise, the moderator M_i be a dummy that takes value 1 if a project is implemented and 0 otherwise, and D_i be a dummy that takes value 1 if the region is the birth region of a leader and 0 otherwise. We have four principal strata corresponding to four possible profiles of $M_i(d)$, which are described in Table D.1.[18] Definition of these principal strata allows to move to discussion of the four assumptions necessary for estimation of the $ATE_{M=1}$, or ATE of birth region among implemented projects.

	$M_i(0) = 0$	$M_i(0) = 1$
$M_i(1) = 0$	Never Implement	Implement if Not Birth Region
$M_i(1) = 1$	Implement if Birth Region	Always Implement

Table D.1: Mediator Value and Treatment Status

Assumption 1 (Project Observability) $Y_i(d, 0) = 0$ *for all* i *and for* $d \in \{0, 1\}$

This assumption states that unplaced (or unobserved) projects do not appear in PAs. Certainly, if we do not observe a project in our data, we will not observe that it is in a PA. This is thus centrally a question of data quality. It is possible that the data around Chinese projects is incomplete, which would violate this assumption. It is unlikely, however, that unobserved projects are likely to be the kinds of projects that end up in PAs. Projects in PAs are generally in infrastructure and other easily observable sectors, while unobserved projects should tend to be smaller and in less observable projects, such as consulting services for the government.

Assumption 2 (Mediator Monotonicity) $M_i(1) \geq M_i(0)$ *for all* i.

18. See Figure 2 in Knox et al.

This assumption rules out the existence of one of our principal strata, *Implement if Not Birth Region*. Intuitively, it is unlikely that projects are implemented if and only if the region is not the leader's birth region; any projects that are implemented in control regions would also be implemented in the treatment region. This is consistent both with the intuition of aid as a political tool and with existing empirical work on the subject (Dreher et al. 2019).

Assumption 3 (Differential Underlying Project Traits) $E[Y_i(d,m) \mid D_i = d', M_i(1) = M_i(0) = 1, X_i = x] \geq E[Y_i(d,m) \mid D_i = d', M_i(1) > M_i(0), X_i = x]$

This assumption states that projects that are implemented regardless of the president's birth region are weakly more likely to be placed in PAs than those that are only implemented in the president's birth region. We believe this is a plausible assumption given that in our theoretical model, China does not choose to implement a project only in a birth region because it can be placed in a PA. PA placement is a secondary process following the implementation decision.

Assumption 4 (Treatment Ignorability)

1. *With respect to potential mediator* $M_i(d) \perp\!\!\!\perp D_i \mid X_i$.
2. *With respect to potential outcomes:* $Y_i(d,m) \perp\!\!\!\perp D_i \mid M_i(0) = m', M_i(1) = m'', X_i$.

This represents a standard assumption of empirical work, indicating that both the treatment (birth region) is as good as randomly assigned to both the mediator (4.1) and the potential outcomes, conditional on some covariate(s) X_i. This assumption necessitates the inclusion of covariates that affect both the likelihood that a region produces a president and the likelihood that a region receives aid projects (and whether those projects are in PAs). We justify a number of controls and includes them in our analyses.

Results

Knox, Lowe, and Mummolo (2020) define the sharp bounds of the $ATE_{M=1}$ the following way.

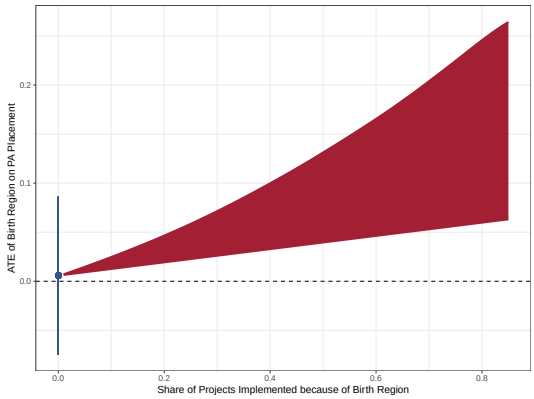
$$\mathbb{E}[\hat{\Delta}] + \rho\mathbb{E}[Y_i|D_1 = 0, M_i = 1](1 - \Pr(D_i = 0|M_i = 1)) \leq ATE_{M=1} \leq$$
$$\mathbb{E}[\hat{\Delta}] + \frac{\rho}{1-\rho}\left(\mathbb{E}[Y_i|D_1 = 1, M_i = 1] - \max\left\{0, 1 + \frac{1}{\rho}\mathbb{E}[Y_i|D_1 = 1, M_i = 1] - \frac{1}{\rho}\right\}\right) \times$$
$$\Pr(D_i = 0|M_i = 1) + \rho\mathbb{E}[Y_i|D_1 = 0, M_i = 1](1 - \Pr(D_i = 0|M_i = 1))$$

Note that $\hat{\Delta}$ refers to the "naive" estimator of the ATE under post-treatment bias defined as $\overline{Y_i|D_i = 1, M_i = 1} - \overline{Y_i|D_i = 0, M_i = 1}$. ρ refers to the share of projects that have only been implemented because they happen the leader's birth region.

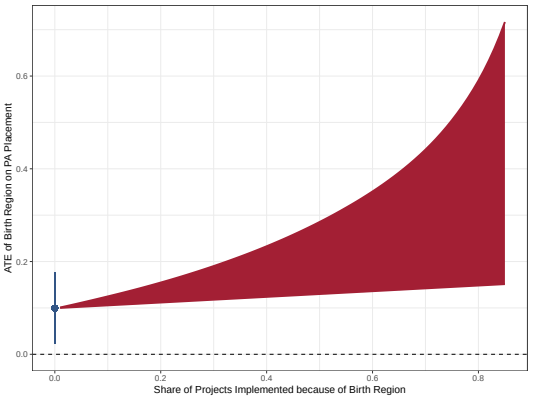
The ATE bounds for our projects are displayed in Figure D.1. The red shaded area represents the range of ATEs possible depending on the share of projects only happening because they are placed in the leader's birth region.

Figure D.1: ATE Bounds of birth region using Knox, Lowe, and Mummolo (2020) method

(a) All



(b) Africa



(c) Outside Africa

